linear algebra

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note
this document needs some refactoring, some of the proofs may be incomplete

1 field

2 matrix

Definition 2.0.1. a **matrix** is a rectangular array or table of numbers, symbols, or expressions, arranged in rows and columns, which is used to represent a mathematical object or a property of such an object

Characteristic 1. a matrix can be constructed over a specific finite/infinite set of numbers for example $\mathbb{Z}$ which is the set of integers, when we say a matrix is over a specific field of numbers we mean that the numbers in the matrix are limited to that field of numbers

Example 2.0.1. consider this matrix over the field $\mathbb{Z}$, where all the numbers are integers

$$\begin{bmatrix} -1 & 1 & -5 & 1 \\ -2 & -3 & 1 & 2 \\ -4 & 0 & 2 & 2 \\ 0 & -1 & -1 & 0 \\ -1 & -5 & 1 & 1 \end{bmatrix}$$

2.1 matrix multiplication

Definition 2.1.1. let $A,\ B$ be matrices
matrix multiplication is possible if and only if the matrices have the dimensions $A_{m\times n},\ B_{n\times m}$
`./matrix-multiplication.gif`

Definition 2.1.2. matrices can be **multiplied by constants**, let A be a matrix and c be a constant, the definition of such multiplication is:

$$A \cdot c = B \iff A_{i,j} \cdot c = B_{i,j}$$

2.2 matrix addition

Definition 2.2.1. matrix addition is defined as the addition of 2 matrices with the same dimensions $A_{m\times n},\ B_{m\times n}$

$$A + B = C \iff A_{i,j} + B_{i,j} = C_{i,j}$$

Example 2.2.1.

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} + \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} = \begin{bmatrix} A_1 + B_1 & A_2 + B_2 \\ A_3 + B_3 & A_4 + B_4 \end{bmatrix}$$

2.3 elementary row operation

2.4 augmented matrix

Definition 2.4.1. for a given system of equation, an **augmented matrix** is the system of equations represented using a matrix

Example 2.4.1.

$$\begin{aligned} 2x + 5y &= 10 \\ 3x + 4y &= 24 \end{aligned} \iff \left[\begin{array}{cc|c} 2 & 5 & 10 \\ 3 & 4 & 24 \end{array}\right]$$

2.5 square matrix

Definition 2.5.1. a **square matrix** is a matrix whose width is the same as its height, i.e. $A_{n\times n}$

2.6 identity matrix

2.7 triangular matrix

2.8 row echelon form

Characteristic 2. once we convert a matrix to its **echelon form**, we would be able to clearly see how many solutions said matrix has
a matrix with no solution has the following echelon shape:

$$\left[\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 5 \end{array}\right]$$

its not possible that $0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 = 5$ so such a matrix has no solution

Characteristic 3. a matrix with a single solution has the following echelon shape:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 5 \end{array}\right]$$

this is the kind of shape we would hope for when reducing a matrix to its echelon form, because the solution is very easy to read which is:

$$\begin{aligned} x_1 &= 2 \\ x_2 &= 0 \\ x_3 &= 5 \end{aligned}$$

Characteristic 4. a matrix with an ∞ number of solutions has the following echelon shape:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

where one of the rows zeros out which makes the third variable in the corresponding system of equations a *free variable* that can take any value, the other 2 variables depend on the value of the free variable

2.9 invertible matrix

2.10 transpose

2.11 matrix similarity

2.12 determinant

see `https://en.wikipedia.org/wiki/Determinant`

Definition 2.12.1. the **determinant** is a scalar value that is a function of the entries of a square matrix, it is nonzero if and only if the matrix is invertible and the linear map represented by the matrix is an isomorphism. the determinant of a matrix A is denoted $\det(A)$, $\det A$, or $|A|$.
in the case of a 2×2 matrix the determinant can be defined as

$$|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

similarly, for a 3×3 matrix A, its determinant is

$$\begin{aligned} |A| = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} &= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ &= aei + bfg + cdh - ceg - bdi - afh \end{aligned}$$

each determinant of a 2×2 matrix in this equation is called a minor of the matrix A. this procedure can be extended to give a recursive definition for the determinant of an $n \times n$ matrix, known as laplace expansion

Characteristic 5. $\det I = 1$ where I is an identity matrix

Characteristic 6. the determinant is multilinear

Characteristic 7. the determinant is alternating

Lemma 1. the determinant is a homogeneous function, i.e.

$$\det(cA) = c^n \det(A)$$

assuming A is an $n \times n$ matrix

Lemma 2. interchanging any pair of columns of a matrix multiplies its determinant by -1. This follows from the determinant being multilinear and alternating

Lemma 3. if the columns of the matrix form a linearly dependant set, the determinant is 0

Lemma 4. adding a scalar multiple of one column to another column does not change the value of the determinant. this is a consequence of multilinearity and being alternative

Lemma 5. if A is a triangular matrix then its determinant equals the product of the diagonal entries:

$$\det A = a_{11}a_{22}\cdots a_{nn} = \prod_{i=1}^{n} a_{ii}$$

Lemma 6. the determinant of the transpose of a matrix equals the determinant of the matrix:

$$\det A^T = \det(A)$$

Lemma 7.

$$\det(AB) = \det A \det B$$

Lemma 8.

$$\det\left(A^{-1}\right) = \frac{1}{\det A} = \det(A)^{-1}$$

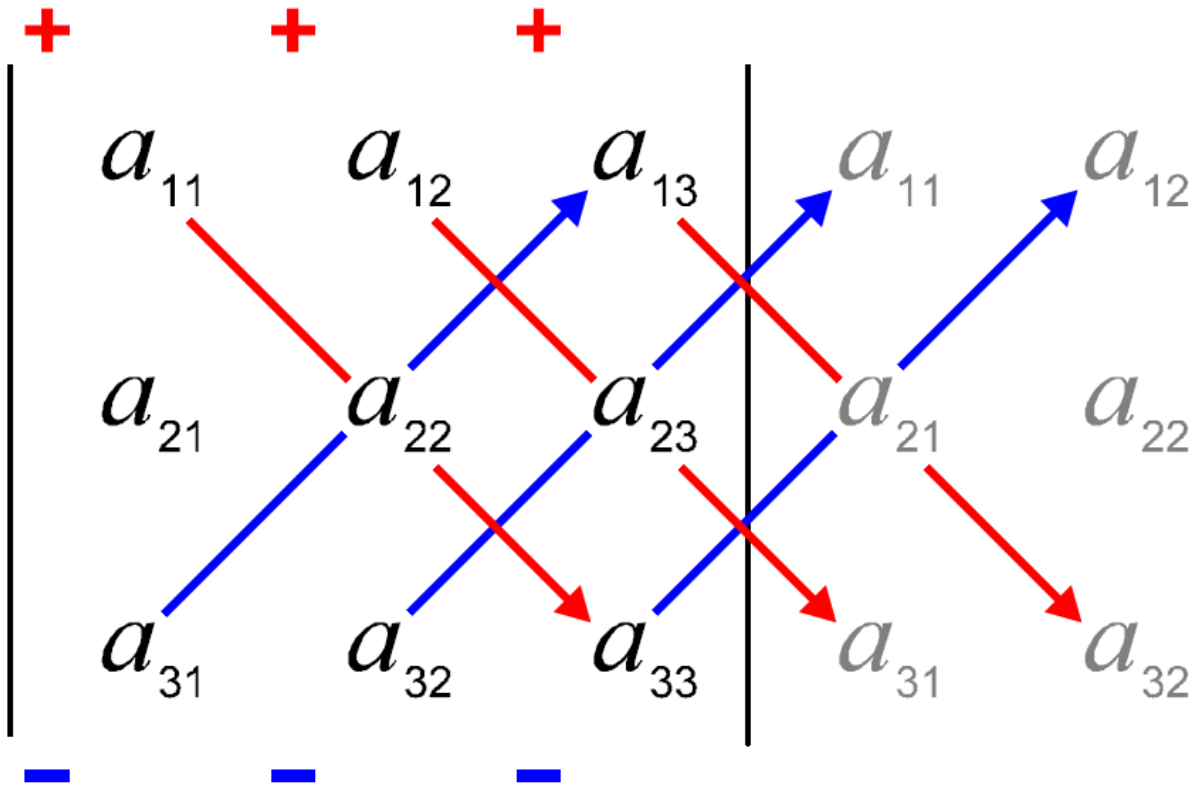
2.12.1 Leibniz formula

see `https://en.wikipedia.org/wiki/Determinant`
the **Leibniz formula** for the determinant of a 3×3 matrix is the following:

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

2.12.2 rule of Sarrus

see `https://en.wikipedia.org/wiki/Rule_of_Sarrus`
the **rule of Sarrus** is a mnemonic of Leibniz formula



$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

2.13 diagonal matrix

Definition 2.13.1. a **diagonal matrix** is a matrix in which the entries outside the main diagonal are all zero; the term usually refers to square matrices. elements of the main diagonal can either be zero or nonzero

2.14 diagonalizable matrix

see `https://textbooks.math.gatech.edu/ila/diagonalization.html`

Definition 2.14.1. a square matrix A is called **diagonalizable** if it is similar to a diagonal matrix, i.e., if there exists an invertible matrix P which we may call the **eigenbasis matrix** or **eigenmatrix** and a diagonal matrix D such that $P^{-1}AP = D$, or equivalently $A = PDP^{-1}$

Characteristic 8. diagonalizable matrices are especially easy for computations, once their eigenvalues and eigenvectors are known, one can raise a diagonal matrix D to a power by simply raising the diagonal entries to that power, and the determinant of a diagonal matrix is simply the product of all diagonal entries, and multiplying diagonal matrices together just multiplies their diagonal entries; such computations generalize easily to

$A = PDP^{-1}$:

$$A^k = T \begin{pmatrix} \lambda_1^k & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2^k & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3^k & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n^k \end{pmatrix} T^{-1}$$

Example 2.14.1.

$$\begin{pmatrix} x_1 & 0 & 0 \\ 0 & x_2 & 0 \\ 0 & 0 & x_3 \end{pmatrix} \begin{pmatrix} y_1 & 0 & 0 \\ 0 & y_2 & 0 \\ 0 & 0 & y_3 \end{pmatrix} = \begin{pmatrix} x_1y_1 & 0 & 0 \\ 0 & x_2y_2 & 0 \\ 0 & 0 & x_3y_3 \end{pmatrix}$$

Example 2.14.2.

$$\begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix}^n = \begin{pmatrix} x^n & 0 & 0 \\ 0 & y^n & 0 \\ 0 & 0 & z^n \end{pmatrix}$$

Example 2.14.3.

$$T^{-1}AT = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$TT^{-1}ATT^{-1} = T \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} T^{-1}$$

$$A = T \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} T^{-1}$$

$$A^2 = T \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} T^{-1}T \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} T^{-1}$$

$$A^2 = T \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}^2 T^{-1}$$

$$A^2 = T \begin{pmatrix} 2^2 & 0 & 0 \\ 0 & 2^2 & 0 \\ 0 & 0 & 1^2 \end{pmatrix} T^{-1}$$

$$A^k = T \begin{pmatrix} \lambda_1^k & 0 & 0 \\ 0 & \lambda_2^k & 0 \\ 0 & 0 & \lambda_3^k \end{pmatrix} T^{-1}$$

Example 2.14.4. any diagonal matrix is diagonalizable because it is **similar** to itself

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} = I_3 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} I_3^{-1}$$

2.14.1 diagonalization theorem

a fundamental question about a matrix is whether or not it is diagonalizable. the following is the primary criterion for diagonalizability. it shows that diagonalizability is an eigenvalue problem.

Definition 2.14.2. an $n \times n$ matrix $A_{n\times n}$ is **diagonalizable** if and only if A has n linearly independant eigenvectors
in this case, $A = CDC^{-1}$ for

$$C = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix} \quad D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

where $v_1, v_2, \ldots, v_n$ are linearly independant eigenvectors, and $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the corresponding eigenvalues, in the same order.

note
the eigenvectors $\lambda_1, \lambda_2, \ldots, \lambda_n$ dont have to be unique, in case they arent, we call that Algebraic multiplicity

2.15 minor

Definition 2.15.1. a **minor** of a matrix A is the determinant of some smaller square matrix, cut down from A by removing one or more of its rows and columns.

2.15.1 first minor

Definition 2.15.2. if A is a square matrix, then the minor of the entry in the ith row and jth column (also called the (i, j) minor) is the determinant of the submatrix formed by deleting the ith row and jth column. this number is often denoted $M_{i,j}$.

2.16 laplace expansion

2.17 conjugate transpose

see `https://en.wikipedia.org/wiki/Conjugate_transpose`

Definition 2.17.1. the **conjugate transpose** of an $m \times n$ matrix $A_{m\times n}$ with complex entries is the $n \times m$ matrix obtained from A by taking the transpose and then taking the complex conjugate of each entry (the complex conjugate of $a + ib$ being $a - ib$, for real numbers a and b). it is often denoted as A^H or A^*.
For real matrices, the conjugate transpose is just the transpose, $A^H = A^T$

2.18 cofactor matrix

Definition 2.18.1. a **cofactor matrix** of a square matrix A, denoted C is the matrix formed by the cofactors of A

Characteristic 9. the cofator matrix is useful to find the adjugate and the inverse of the given matrix.

2.18.1 cofactor

Definition 2.18.2. the (i, j) **cofactor** of a matrix, often denoted $C_{i,j}$ is obtained by multiplying the first minor $M_{i,j}$ by $(-1)^{i+j}$.

2.19 adjugate matrix

Definition 2.19.1. the **adjugate** of a square matrix is the transpose of its cofactor matrix.

Characteristic 10. the product of a matrix with its adjugate gives a diagonal matrix whose diagonal entries are the determinant of the original matrix. consequently, dividing this product matrix by the determinant produces the identity matrix of that size, which is just a diagonal matrix with all the diagonal entries equal to one. The multiplicative inverse of a square matrix can therefore be found by dividing its adjugate by its determinant, since multiplying this by the original matrix gives the identity.

Characteristic 11. for any $n \times n$ matrix A, elementary row operation show that adjugates have the following properties:

Lemma 9.

$$\operatorname{adj} 0 = 0$$

Lemma 10.

$$\operatorname{adj} I = I$$

Lemma 11.

$$\operatorname{adj}\left(A^T\right) = \operatorname{adj}(A)^T$$

Lemma 12. if A is invertible, or equivalently, if $|A| \neq 0$

$$A^{-1} = \frac{\operatorname{adj} A}{|A|}$$

Entailment 1. $\operatorname{adj} A$ is invertible and its inverse is $(\det A)^{-1} A$

Entailment 2.

$$\operatorname{adj} A^{-1} = \operatorname{adj}(A)^{-1}$$

Lemma 13.

$$\operatorname{adj} AB_{n\times n} = \operatorname{adj} B_{n\times n} \operatorname{adj} A$$

Entailment 3.

$$\operatorname{adj} A^k = \operatorname{adj}(A)^k$$

2.20 matrix from the perspective of vector

2.20.1 row space

Definition 2.20.1. the **row space** of a matrix A, denoted by $R(A)$, is the span of its rows

> **note**
> here we are basically looking at the matrix as a column of rows where each row represents a vector

$$\underbrace{\begin{bmatrix} 4 & 1 & 5 & 11 \\ 3 & 2 & 7 & 9 \\ 2 & 3 & 9 & 7 \\ 1 & 4 & 11 & 5 \end{bmatrix}}_{\text{every vector is of size 4}} \left.\vphantom{\begin{matrix} 4 \\ 3 \\ 2 \\ 1 \end{matrix}}\right\} \text{4 row vectors}$$

2.20.2 column space

Definition 2.20.2. the span of the columns of a matrix A, denoted by $C(A)$, is called the **range** or the **column space** of the matrix. the row space and the **column space** of a matrix always have the same dimension
generally when referring to a span of a matrix we refer to the **column space**
here we are basically looking at the matrix as a row of columns where each column represents a vector

$$\underbrace{\left.\begin{bmatrix} 4 & 1 & 5 & 11 \\ 3 & 2 & 7 & 9 \\ 2 & 3 & 9 & 7 \\ 1 & 4 & 11 & 5 \end{bmatrix}\right\}}_{\text{4 column vectors}} \text{every vector is of size 4}$$

note

please note that elementary row operations change the column space (unlike the row space which doesnt change) so after reducing a matrix you would have to go back to the original matrix and pick the corresponding columns from there not from the matrix you applied row operations to

Example 2.20.1. say $A_{m\times n} \in M_{m\times n}(\mathbb{R})$ then $C(A) \leq \mathbb{R}^m$ and $R(A) \leq \mathbb{R}^n$

Characteristic 12. consider the transposition of A, $(A^t)_{n\times m}$, we know $C(A^t) = R(A)$ and $R(A^t) = C(A)$

Characteristic 13. we can describe matrix multiplication using vectors and the concept of linear combinations if $A \cdot B = C$ then the column C^j (column j of C) is a linear combination of the columns of A using B^j as the set of coefficients

Example 2.20.2.

$$\begin{bmatrix} 3 & 1 \\ 2 & 0 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 3 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 & 4 \\ 2 & 0 & 2 \\ 7 & 4 & 3 \end{bmatrix}$$

you might notice that:

$$1 \cdot \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 7 \end{bmatrix}$$

here the output vector is a linear combination of the 2 vectors and $1, 3$ are the coefficients

Entailment 4. the columns of A span the columns of AB

$$C(AB) \leq C(A)$$

the rows of B span the rows of AB

$$R(AB) \leq R(B)$$

Entailment 5.

$$\dim C(AB) \leq \dim C(A)$$
$$\dim R(AB) \leq \dim R(B)$$

Entailment 6. given $A_{m\times n}$

$$C(A) \leq \mathbb{R}^m$$
$$R(A) \leq \mathbb{R}^n$$
$$\dim C(A) \leq m$$
$$\dim R(A) \leq n$$
$$\dim C(A) \leq n$$
$$\dim R(A) \leq m$$
$$\dim C(A) \leq \min\{n, m\}$$
$$\dim R(A) \leq \min\{n, m\}$$

Lemma 14. if L is an invertible matrix then:

$$\dim R(LB) \leq \dim R(B)$$

Proof.

$$B = L^{-1}LB$$
$$R(B) = R(L^{-1}(LB)) \leq R(LB) \leq R(B)$$

□

Entailment 7.

$$R(LB) = R(B)$$

Lemma 15. if L is an invertible matrix then

$$C(BL) \leq C(B)$$

Proof.

$$B = BLL^{-1}$$
$$C(B) = C((BL)L^{-1}) \leq C(BL)$$

□

Entailment 8. $R(A) = R(A')$ where A is the matrix in its reduced echelon form

Lemma 16. given the matrix $A_{m\times n}$

$$\dim C(A) \leq \dim R(A)$$

Proof. let $\dim R(A) = k$ and let $b_1, b_2 \dots b_k$ be a basis of $R(A)$
we construct a matrix B

$$B = \begin{bmatrix} \cdots b_1 \cdots \\ \cdots b_2 \cdots \\ \vdots \\ \cdots b_n \cdots \end{bmatrix}_{k \times n}$$

we know $k \leq m$
there exists Δ such that $\Delta \cdot B = A$ because the rows of A are spanned by the rows of B

$$C(\Delta B) \leq C(\Delta)$$

therefore

$$\dim C(A) = \dim C(\Delta B) \leq \dim(C(\Delta)) \leq k = \dim R(A)$$

we substitute A inplace of A^t in the lemma and we get:

$$\begin{aligned} \dim C(A^t) &\leq \dim R(A^t) \\ \dim R(A) &\leq \dim C(A) \end{aligned}$$

therefore

$$\dim C(A) = \dim R(A)$$

□

2.20.3 rank

Definition 2.20.3. the **rank** of a given matrix A is the dimension of its row space which is also equal to the dimension of its column space

$$\operatorname{rank}(A) = \dim C(A) = \dim R(A)$$

> **note**
> if the matrix is given, the **rank** can be simply found by checking how many linearly independant rows (or columns) the matrix has

Characteristic 14. given the matrix $A_{m \times n}$, $\ker A = \{v \mid A\vec{v} = \vec{0}\}$, $\ker A \leq \mathbb{R}^n$, given A' is A in its reduced echelon form then $\dim \ker A =$ number of free variables in A' and $\operatorname{rank} A = \dim R(A) = \dim R(A')$

Lemma 17. for every matrix $A_{m \times n}$:

$$\overbrace{\dim \ker A}^{\text{number of free variables}} + \underbrace{\operatorname{rank} A}_{\text{number of basic variables}} = n$$

Question 1. given the matrices A, B such that $A_{n\times k} \cdot B_{k\times m} = 0_{n\times m}$, prove that $\ker A \leq C(B)$

Characteristic 15. To find a basis for the span of a set of vectors, write the vectors as rows of a matrix and then row reduce the matrix. The span of the rows of a matrix is called the row space of the matrix. The dimension of the row space is the rank of the matrix.

2.20.4 kernel

Definition 2.20.4. the **kernel** of a matrix or linear transformation A is a set of all vectors V such that for every $v \in V$ the expression $Av = 0$ holds true in the case of matrix and $A(v) = 0$ in the case of linear transformation

Characteristic 16. the kernel of a matrix doesnt change when an elementary row operation is applied to the matrix

2.20.5 image

Definition 2.20.5. The **image** of a linear transformation or matrix is the span of the column vectors of the linear transformation. (think of it as what you can get from applying the linear transformation or multiplying the matrix by a vector.) it can be written as $\mathrm{Im}(A)$

3 vector

anything can be a vector, so long as we define sensible addition and scalar multiplication operations that abide by the axioms of a vector space
a **vector** can be represented using its components or using length and direction

3.1 magnitude

Definition 3.1.1. the **magnitude** of a vector A is its length, denoted by $|A|$ or $\|A\|$, it can be calculated from the square root of the total of the squares of of the individual vector components

3.2 unit vector

Definition 3.2.1. a vector whose magnitude is equal to 1, laballed by a caret, e.g. $\hat{\mathbf{i}}$

Lemma 18. $\hat{\mathbf{i}} = \frac{i}{|i|}$
where i is some vector and $\hat{\mathbf{i}}$ is the corresponding unit vector

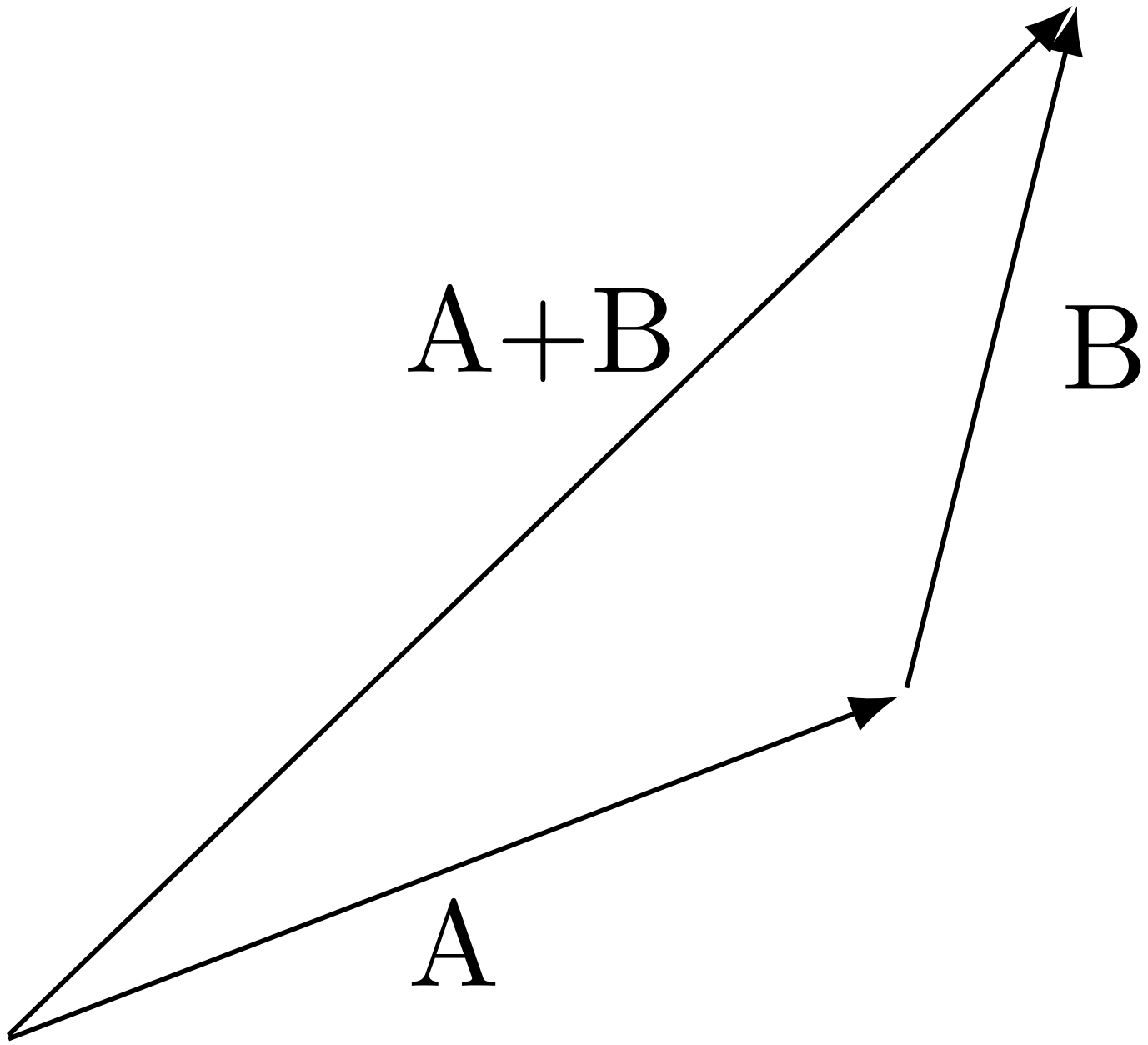
3.3 vector scalar multiplication

Lemma 19. if we multiply a vector A by a positive scalar b, the result is a new vector $C = bA$. The vector C is parallel to A, and its length is b times greater, thus $\hat{C} = \hat{A}$ (unit vectors) and $|C| = b|A|$

Lemma 20. the result of multiplying a vector by -1 is a new vector opposite in direction to the original vector multiplication of a vector by a negative scalar evidently can change both the magnitude and the direction sense

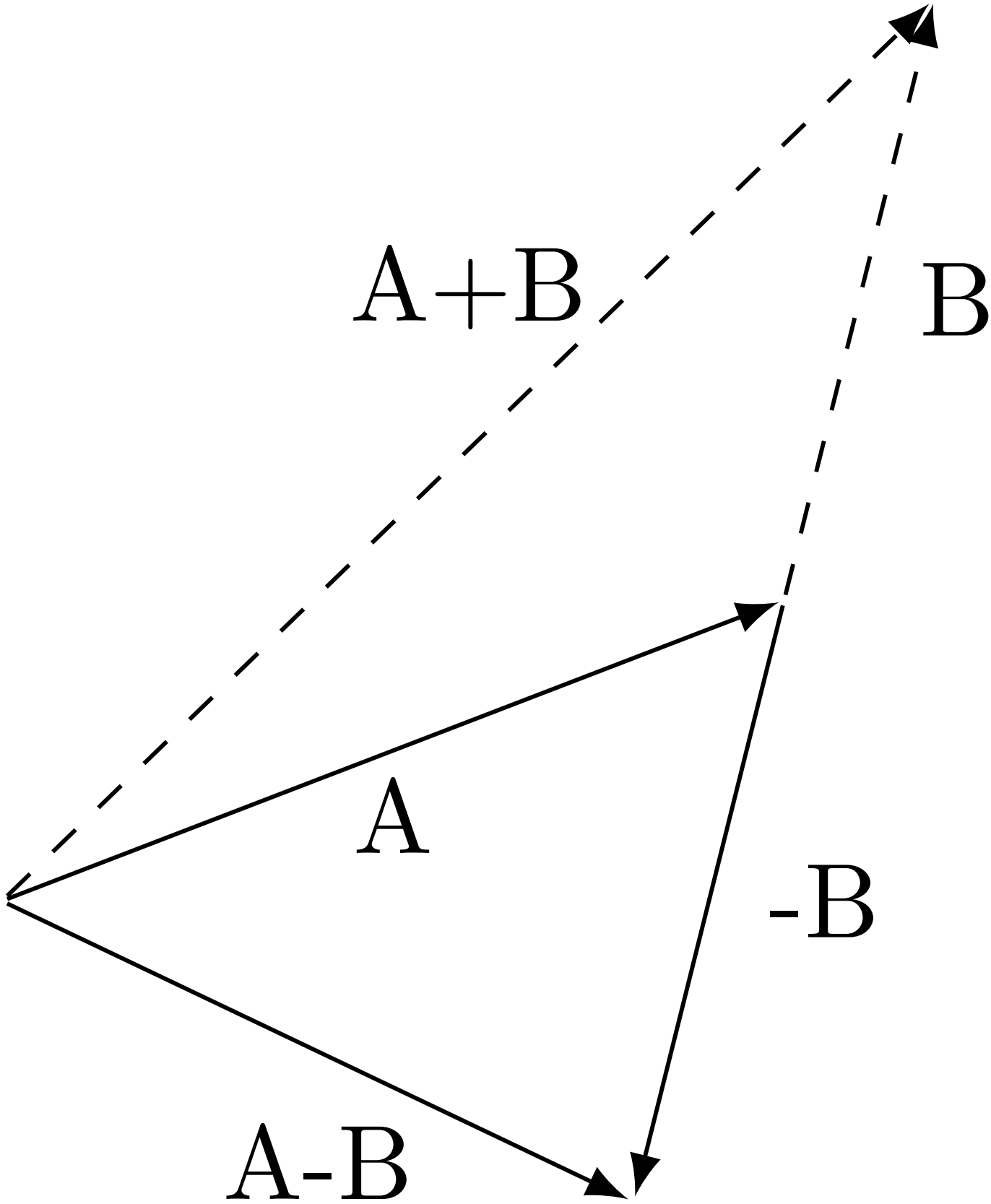
3.4 vector addition and subtraction

the addition of 2 vectors is simply described in the following drawing:



the rule is: to add B to A, place the tail of B at the head of A. the sum is a vector from the tail of A to the head of B

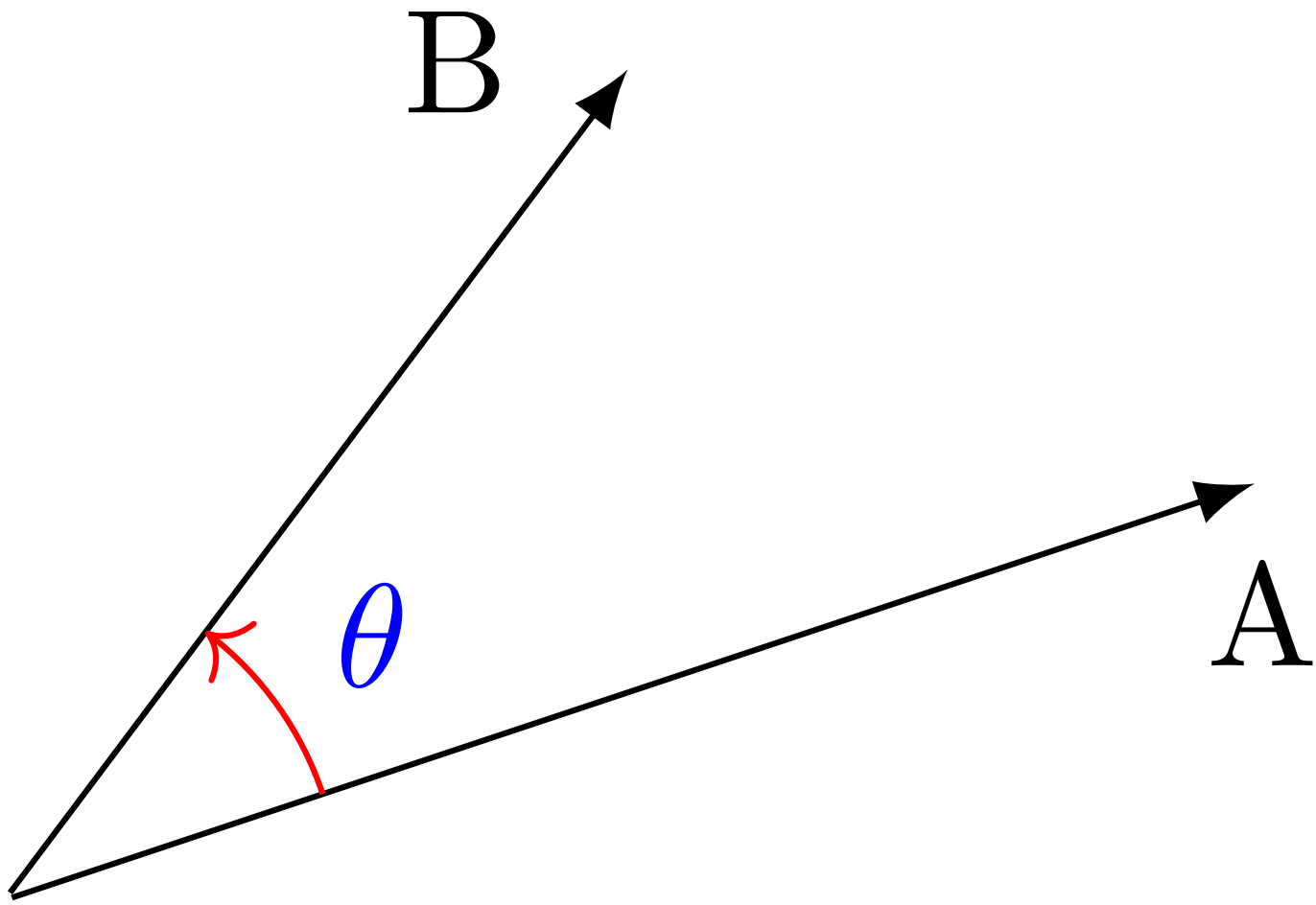
since $A - B = A + (-B)$, in order to subtract B from A we can simply multiply it by -1 and then add, so in a sense, we're "rotating" B and adding it to A to get $A - B$:



vector addition is **commutative**, associative and distributive

3.5 dot product

Definition 3.5.1. the scalar product of 2 vectors A and B is defined as $A \cdot B = |A||B| \cos\theta$ where θ is the angle between A and B when they are drawn tail to tail (to eliminate ambiguity, θ is always taken as the angle smaller than π):



when the vectors are in the form of a list of components, e.g. (x, y, z) then the dot product is the sum of the products of corresponding components, i.e. given two vectors $A = (A_1, \ldots, A_n), B = (B_1, \ldots, B_n)$, their dot product is $A \cdot B = A_1B_1 + \cdots + A_nB_n$

Question 2. find a unit vector in the xy plane which is perpendicular to $A = (3, 5, 1)$

Answer 2.0.1. we denote the perpendicular vector by $B = (B_x, B_y, B_z)$, since B is in the xy plane, $B_z = 0$, for B to be perpendicular to A, we have $A \cdot B = 0$ because $\cos 90 = 0$, so:

$$\begin{aligned} A \cdot B &= 3B_x + 5B_y \\ &= 0 \end{aligned}$$

hence $B_y = -\frac{3}{5}B_x$, however, B is a unit vector, which means that $\sqrt{B_x^2 + B_y^2} = 1$
combining these gives $B_x^2 + \frac{9}{25}B_x^2 = 1$, or $B_x = \sqrt{\frac{25}{34}} = \pm 0.857$ and $B_y = -\frac{3}{5}B_x = \pm 0.514$
the ambiguity in sign of B_x and B_y indicates that B can point along a line perpendicular to A in either of two directions

3.6 cross product

unlike the dot product which gives us a scalar as a result, the **cross product** results in a vector, so we get a magnitude and a direction that represent the vector
the cross product is defined by the formula:
$A \times B = \|A\| \, \|B\| \sin(\theta) n$
where θ is the angle between A and B when they are drawn tail to tail (to eliminate ambiguity, θ is always taken as the angle smaller than π)
when we draw A and B tail to tail, they determine a plane, we define n to be a unit vector that is perpendicular to that plane, which means $A \times B$ is in that same direction as the unit vector
imagine a right hand coordinate system with A and B in the xy plane, A lies on the x axis and B lies toward

the y axis, A, B and n should form what is called a right hand triple such that n would be represented by the thumb
a result of our definition of the cross product is that:
$B \times A = -A \times B$
and so obviously cross product isnt commutative but rather anticommutative

3.7 vector space

Definition 3.7.1. let V be a non-empty set of vectors containing numbers from the field $\mathbb{F}$, consider the 2 operations addition and scalar multiplication, $V_{\mathbb{F}}$ is a **vector space** only if it abides by the following axioms:
addition axioms:

- addition closure: for every $\vec{v}_1, \vec{v}_2 \in V$ we have $\vec{v}_1 + \vec{v}_2 \in V$
- associative addition: for every $\vec{v}_1, \vec{v}_2, \vec{v}_3 \in V$ we have $(\vec{v}_1 + \vec{v}_2) + \vec{v}_3 = \vec{v}_1 + (\vec{v}_2 + \vec{v}_3)$
- commutative addition: $\vec{v}_1 + \vec{v}_2 = \vec{v}_2 + \vec{v}_1$
- zero vector: $\vec{0} \in V$ so that $\vec{0} + \vec{v} = \vec{v}$
- negative vector: for every $\vec{v} \in V$ there exists $-\vec{v}$ so that $\vec{v} + (-\vec{v}) = \vec{0}$

multiplication axioms:

- multiplication closure: for every $\vec{v} \in V$ and $a \in \mathbb{F}$ we have $a \cdot \vec{v} \in V$
- associative multiplication: for every $\vec{v} \in V$ and $a, b \in \mathbb{F}$ we have $(a \cdot b) \cdot \vec{v} = a \cdot (b \cdot \vec{v})$
- identity vector: for every $\vec{v} \in V$ we have $\vec{1} \cdot \vec{v} = \vec{v}$
- identity law: for every $\vec{v} \in V$ we have $1 \cdot \vec{v} = \vec{v}$
- first distributive law: for every $a, b \in \mathbb{F}$ and $\vec{v} \in V$ we can have $(a + b) \cdot v = a \cdot v + b \cdot v$
- second distributive law: for every $a \in \mathbb{F}$ and $\vec{v}_1, \vec{v}_2 \in V$ we can have $a \cdot (\vec{v}_1 + \vec{v}_2) = a \cdot \vec{v}_1 + a \cdot \vec{v}_2$

Example 3.7.1. $v = \{\overline{0}\}$ over some field $\mathbb{F}$

Example 3.7.2. over some $n \in \mathbb{N}$ and some field $\mathbb{F}$:

$$v = \{(x_1, x_2, x_3, \dots x_n) \mid x_i \in \mathbb{F}, i = 1, \dots, n\}$$

the definition of summation would be:

$$v = (v_1, v_2, \dots, v_n) \in V$$
$$u = (u_1, u_2, \dots, u_n) \in V$$
$$u \oplus v = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$

and for some $a \in \mathbb{F}$ the definition of multiplication would be:

$$a \odot v = (a \cdot v_1, a \cdot v_2, \ldots, a \cdot v_n)$$

Example 3.7.3. this is an example of the 1st addition axiom
let $\mathbb{F}$ be a field
this describes all the polynomials of x over the field $\mathbb{F}$

$$\mathbb{F}[x] = \sum_{i=0}^{n} a_i \cdot x^i \text{ where } a_i \in \mathbb{F}$$

let $\mathbb{F} = \mathbb{R}$ as an example
the addition of 2 polynomials is as follows:

$$\begin{aligned}
(3x^5 + 2x^3 + x + 1) + (7x^4 + 2x + 3) &= \\
(3x^5 + 0x^4 + 2x^3 + 0x^2 + x + 1) + (0x^5 + 7x^4 + 0x^3 + 0x^2 + 2x + 3) &= \\
3x^5 + 7x^4 + 2x^3 + 0x^2 + 3x + 4 &= \\
3x^5 + 7x^4 + 2x^3 + 3x + 4 &
\end{aligned}$$

the symbolic process of addition can be described as follows:
let $f, g \in \mathbb{F}[x]$ so there exist the polynomial degrees $m \leq n$

$$\begin{aligned}
f(x) &= \sum_{i=0}^{m} a_i \cdot x^i, \ a_i \in \mathbb{F} \\
g(x) &= \sum_{i=0}^{n} b_i \cdot x^i, \ b_i \in \mathbb{F} \\
f(x) \oplus g(x) &= \sum_{i=0}^{m} (a_i + b_i) \cdot x^i + \sum_{i=m+1}^{n} b_i \cdot x^i
\end{aligned}$$

and the symbolic process of constant multiplication is defined as:

$$a \cdot f(x) = \sum_{i=0}^{m} a \cdot a_i \cdot x^i$$

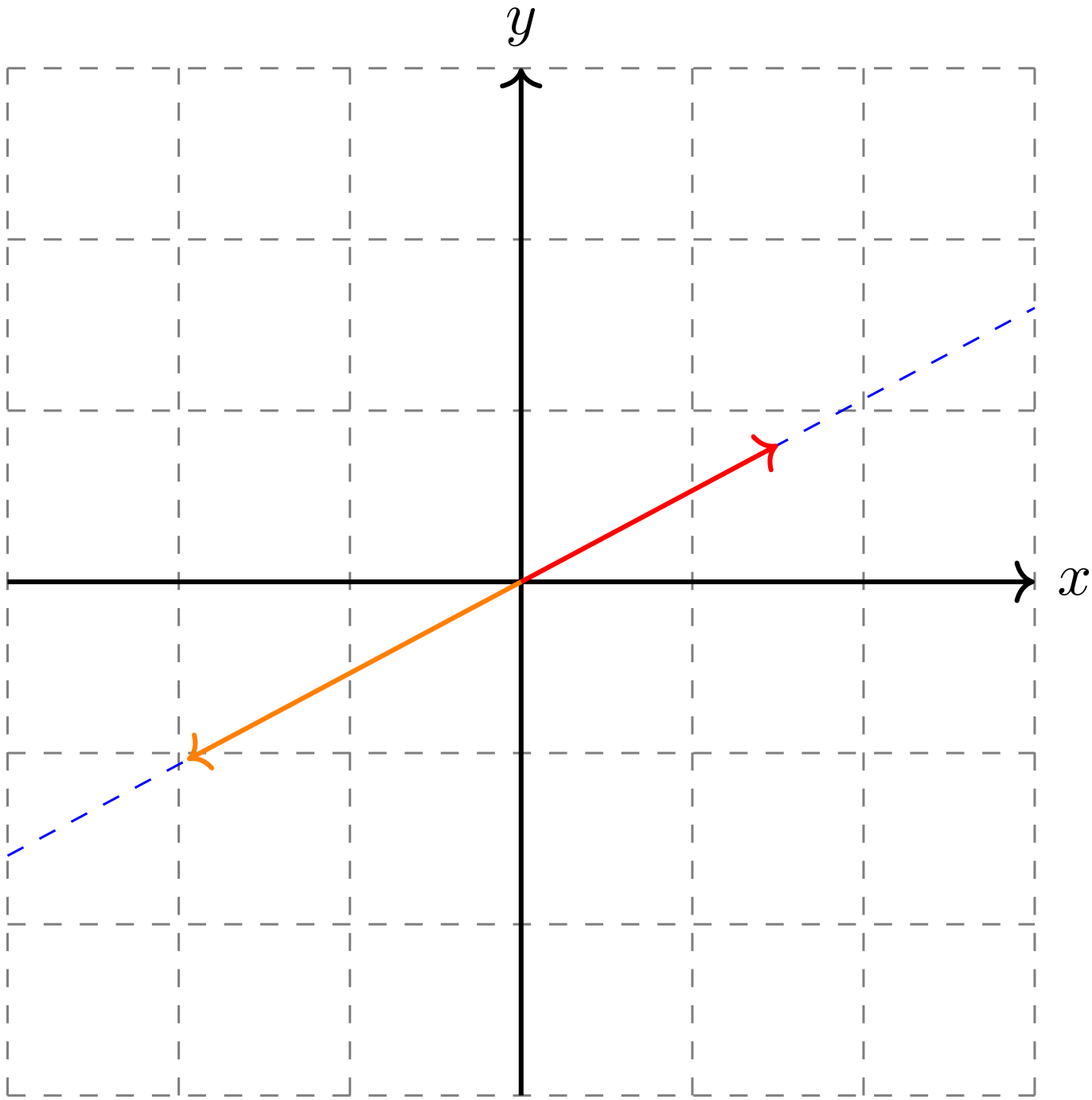
as for the degrees of the resulting polynomials after multiplication/addition:
if $n > m$:

$$\begin{aligned}
\deg(f) &= n \\
\deg(g) &= m \\
\deg(f + g) &= n
\end{aligned}$$

if $n = m$:

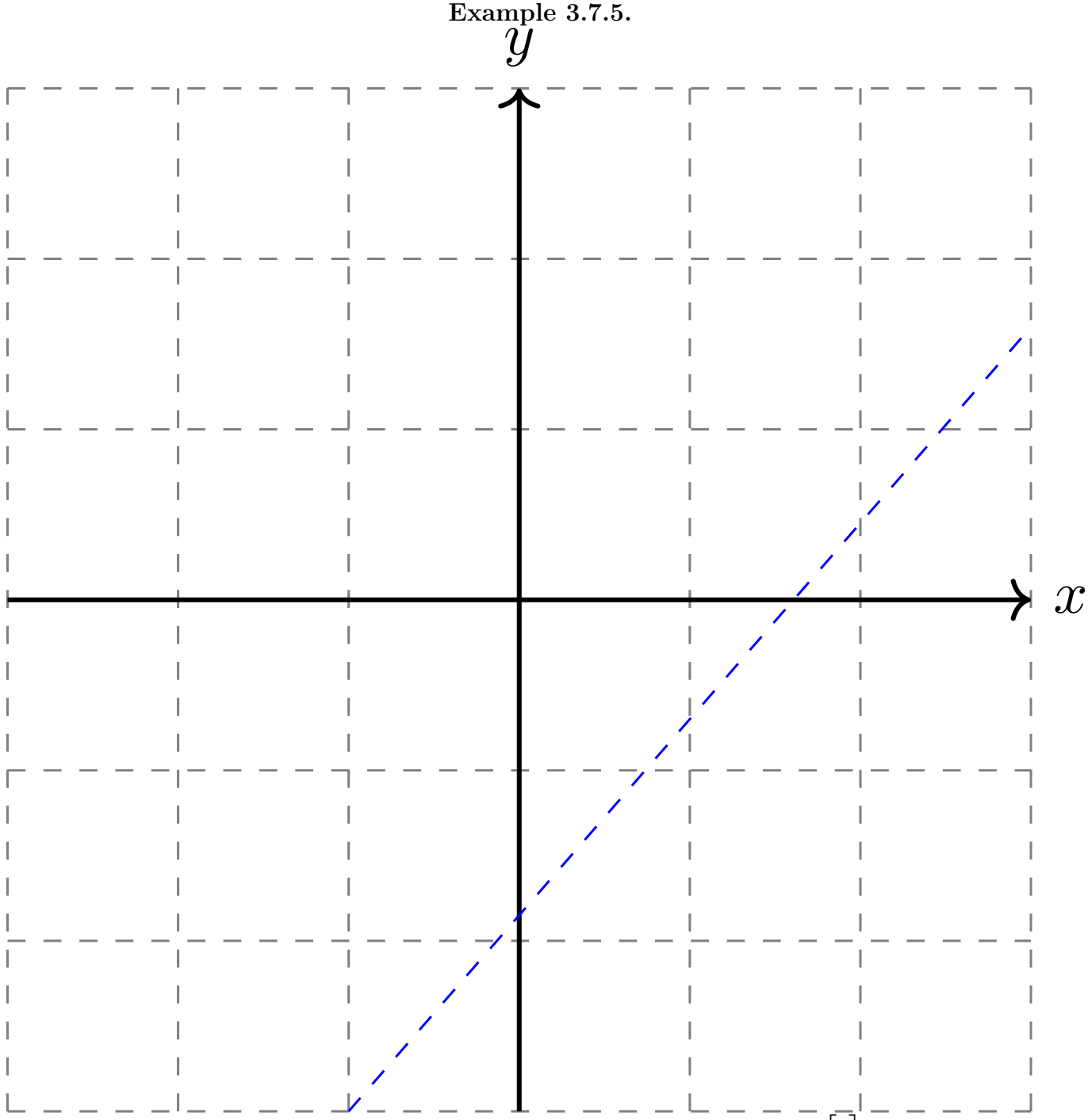
$$\deg(f + g) = \text{cant be predetermined, depends on the polynomials}$$

Example 3.7.4. a vector space could look something like this:



all the vectors that lie on the blue line represent a vector space, because the multiplication of a line would just make it longer (or shorter) it wouldnt make it move out of the blue line, and addition of any 2 vectors that lie on the blue line would also result in a longer (or shorter) vector that lies on the same line which expands across the entire 2d space

Example 3.7.5.



this line doesnt represent a vector space because it doesnt contain the vector $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Lemma 21. reduction law
for every $y, v, w \in V$

$$u + w = v + w \implies u = v$$

Proof.

$$\begin{aligned} (u+w)+(-w) &= (v+w)+(-w) && \text{5th addition axiom (negative vector)} \\ u+(w+(-w)) &= v+(w+(-w)) && \text{2nd addition axiom (associative property)} \\ u+0 &= v+0 \\ u &= v \end{aligned}$$

□

Lemma 22. for $v \in V$:

$$0 \cdot v = \vec{0}$$

Lemma 23. for $a \in \mathbb{F}$

$$a \cdot \vec{0} = \vec{0}$$

Proof.

$$\begin{aligned} a \cdot \vec{0} &= a \cdot (\vec{0} + \vec{0}) \\ a \cdot \vec{0} &= a \cdot \vec{0} + a \cdot \vec{0} \\ \vec{0} + a \cdot \vec{0} &= a \cdot \vec{0} + a \cdot \vec{0} \\ \vec{0} &= a \cdot \vec{0} \end{aligned}$$

□

Lemma 24. $v \in V \implies (-1) \cdot v = -v$

Lemma 25. $v_1 = -v \wedge v_2 = -v \implies v_1 = v_2$

Lemma 26. for $a \in \mathbb{F}$ and $v \in V$

$$a \cdot v = \vec{0} \iff (a = 0 \text{ or } v = \vec{0})$$

3.7.1 vector subspace

Definition 3.7.2. let $V_{\mathbb{F}}$ be a vector space
if $U \subseteq V_{\mathbb{F}}$ we say U is a **subspace** of V and we write $U \leq V$ only if the following is true:

- $\vec{0} \in U$
- for $u_1, u_2 \in U$ its also true that $u_1 + u_2 \in U$ (addition closure)
- for $u \in U,\ a \in F \implies a \cdot u \in U$ (scalar multiplication closure)

Lemma 27. if $U \leq V$ then $U_{\mathbb{F}}$ is a vector space

Proof. let $u_1, u_2 \in U$ then $u_1 + u_2 \in U$, let $u_1, u_2, u_3 \in U$ then $(u_1 + u_2) + u_3 = u_1 + (u_2 + u_3)$
to be continued..

□

Characteristic 17. strictly speaking, a **subspace** is a vector space included in another larger vector space. therefore, all properties of a vector space, such as being closed under addition and scalar multiplication still hold true when applied to the subspace.

Example 3.7.6. let the set of vectors $V_{\mathbb{R}} = \mathbb{R}^2$, consider the following set:

$$V = \{(x, y) \mid x + y = 0\}$$

by the definition of this set, a vector $\vec{v} = (x, y)$ is in the set V only if $x + y = 0$
for $V_{\mathbb{R}}$ to be a subspace of $\mathbb{R}^2$, it has to meet 3 conditions:
condition 1: $\vec{0} \in V$, in this case $\vec{0} = (0, 0)$

$$0 + 0 = 0 \checkmark$$

condition 2: addition closure, meaning if $\vec{v}_1, \vec{v}_2 \in V$ then $\vec{v}_1 + \vec{v}_2 \in V$

$$\begin{aligned} \vec{v}_1 &= (x_1, y_1) \\ \vec{v}_2 &= (x_2, y_2) \\ \vec{v}_3 &= (x_1 + x_2, y_1 + y_2) \\ (x_1 + x_2) + (y_1 + y_2) &= (x_1 + y_1) + (x_2 + y_2) = 0 \checkmark \end{aligned}$$

condition 3: multiplication closure, meaning if $\vec{v} \in V, k \in \mathbb{R}$ then $k \cdot v \in V$

$$\begin{aligned} v &= (x, y) \\ k \cdot v &= (kx, ky) \\ kx + ky &= k(x + y) = 0 \checkmark \end{aligned}$$

3.7.2 linear combination

Definition 3.7.3. let $V_{\mathbb{F}}$ be a vector space, let $A = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\} \subseteq V_{\mathbb{F}}$ and $x_1, x_2, \ldots, x_n \in \mathbb{F}$, then $x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_n\vec{v}_n$ is a **linear combination** of elements of A
although the term **linear combination** is somewhat ambiguous as it can refer to either the combination of vectors and coefficients itself or (in most cases) to its value, i.e. $v = x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_n\vec{v}_n$ means the vector v is a linear combination of the vectors in the set A

3.7.3 span

Definition 3.7.4. the **span** of a given set of vectors is the collection of all the possible linear combinations of the vectors of the set, i.e.:
let $V_{\mathbb{F}}$ be a vector space, let $A = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\} \subseteq V_{\mathbb{F}}$ and $x_1, x_2, \ldots, x_n \in \mathbb{F}$, then $\text{span}(\{\vec{v}_1, \ldots, \vec{v}_n\}) = \{x_1\vec{v}_1, x_2\vec{v}_2, \ldots, x_n\vec{v}_n\}$

Lemma 28. let $A = \{\vec{v_1}, \ldots, \vec{v_n}\} \subseteq V_{\mathbb{F}}$, then $\text{span}(A) = \text{span}\{\vec{v}_1, \ldots, \vec{v}_n\} \leq V_{\mathbb{F}}$

Proof. let $\vec{v} \in \text{span}(A)$, $\vec{x} \in \mathbb{F}$

$$\begin{aligned} x_i v_i &\in V && \text{multiplication closure of } V \\ \vec{v} = \sum_{i=1}^{n} x_i v_i &\in V && \text{addition closure of } V \\ \text{span}(A) &\subseteq V && \text{because every } \vec{v} \in \text{span}(A) \text{ is also } \vec{v} \in V \end{aligned}$$

we have proven that $\text{span}(A)$ is a subset of V, now with that out of the way we need to check the 3 conditions that need to be satisfied for a subset to be a subspace

condition 1: $\vec{0} \in \text{span}(A)$

$$\vec{x} = \vec{0} \implies \sum_{i=1}^{n} 0 \cdot v_i = \vec{0} \in \text{span}(A)$$

condition 2: addition closure
let $\vec{v}, \vec{w} \in \text{span}(A)$ so there exist $\vec{x}, \vec{y} \in \mathbb{F}$ so that:

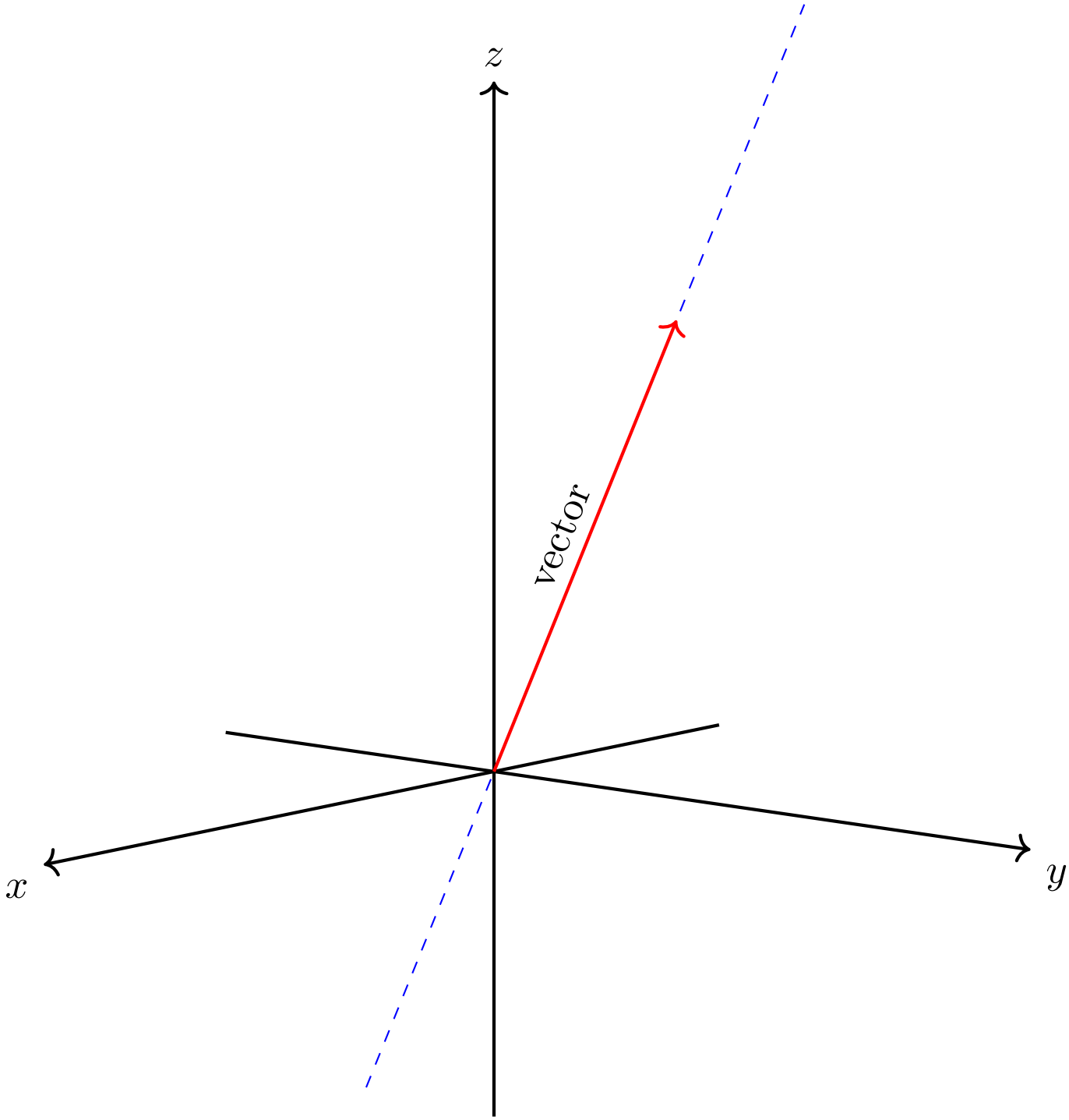
$$\begin{aligned}
\vec{v} &= x_1\vec{v}_1 + x_2\vec{v}_2 + \ldots + x_n\vec{v}_n \\
\vec{w} &= y_1\vec{v}_1 + y_2\vec{v}_2 + \ldots + y_n\vec{v}_n \\
\vec{v} + \vec{w} &= (x_1\vec{v}_1 + \ldots + x_n\vec{v}_n) + (y_1\vec{v}_1 + \ldots + y_n\vec{v}_n) \\
&= \underbrace{(x_1 + y_1)\vec{v}_1}_{(x_1+y_1)\vec{v}_1 \in V} + \ldots + \underbrace{(x_n + y_n)}_{x_n+y_n \in \mathbb{F}}\vec{v}_n \in \text{span}(A)
\end{aligned}$$

condition 3: multiplication closure
let $\vec{v} \in \text{span}\{\vec{v}_1 \ldots \vec{v}_n\}, a \in \mathbb{F}$ so there exist $x_1 \ldots x_n \in \mathbb{F}$, such that $\vec{v} = \sum_{k=1}^{n} x_k\vec{v}_k$

$$a\vec{v} = a\sum_{k=1}^{n} x_k\vec{v}_k = \underbrace{(ax_1)}_{ax_1 \in \mathbb{F}}\vec{v}_1 + (ax_2)\vec{v}_2 + \ldots + (ax_n)\vec{v}_n \in \text{span}\{\vec{v}_1, \ldots, \vec{v}_n\}$$

$\square$

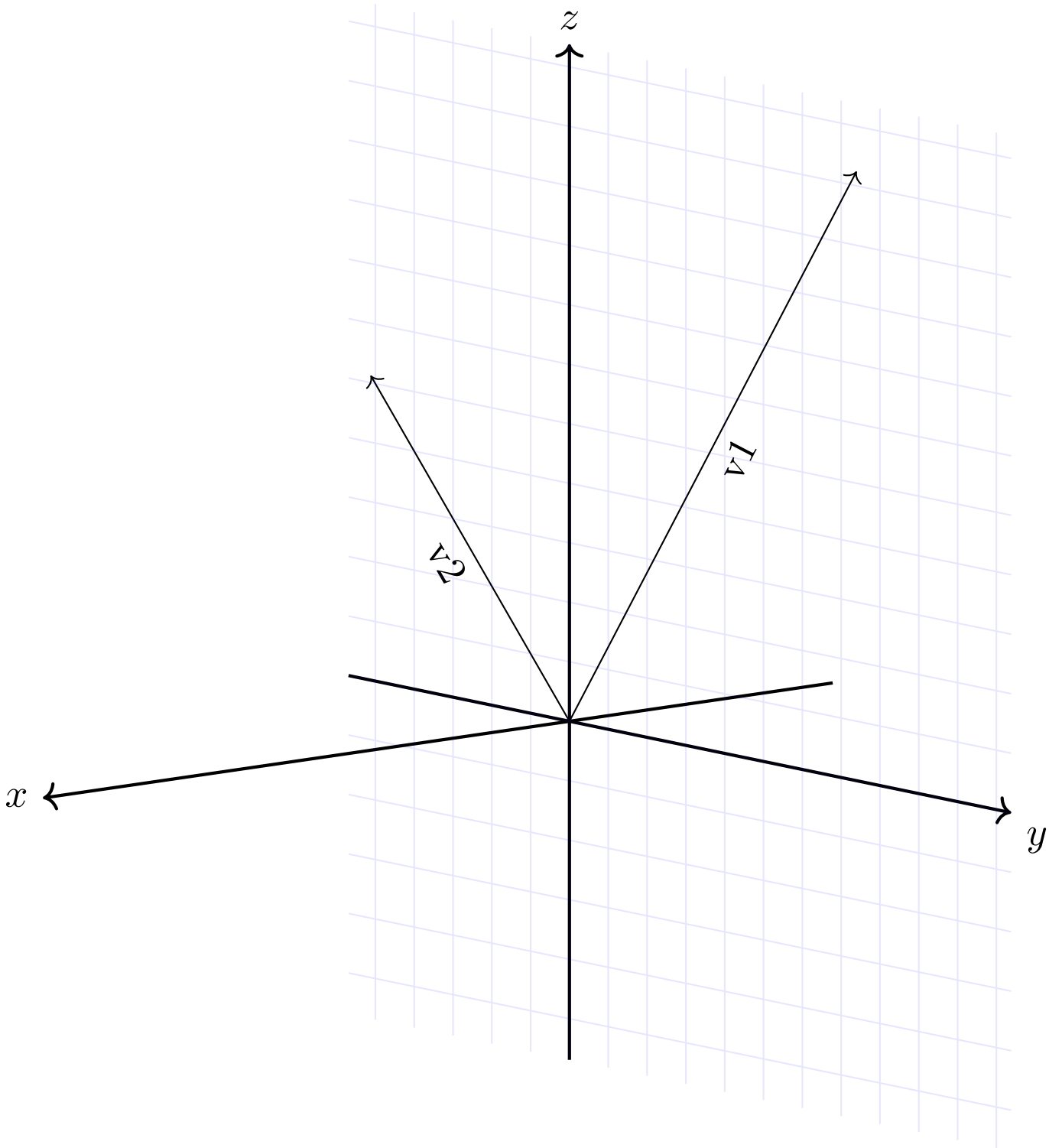
Example 3.7.7. by definition of **span**, a span of a set of vectors is the collection of all the possible linear combinations using said vectors
the span of a single vector would just be its own line expanded across the entire 3d region, because we cant reach other dimensions by constant multiplication or by addition of the vector to itself



the span of 2 **linearly independant** vectors is however more interesting because it would be visualized as a grid across the 3d space

for simplicity, we take the relatively simple vectors $\begin{bmatrix} 0 \\ 1.3 \\ 1.8 \end{bmatrix}$, $\begin{bmatrix} 0 \\ -0.9 \\ 0.9 \end{bmatrix}$ that lie on the $x = 0$ grid which covers the y and z axes

z
v1
v2
x
y



spanning set

Definition 3.7.5. consider the vector space V, let $w \subseteq V_{\mathbb{F}},\ |w| \in \mathbb{N},\ \text{span}(w) = V$, then w is a **spanning set** of V and V is a **finitely spanned set**

Example 3.7.8.

- $\text{span}\{(1,0,0),(0,1,0),(0,0,1)\} = \mathbb{R}^3$

- $\text{span}\{(1,0,0),(0,1,0),(1,1,0)\} \neq \mathbb{R}^3$ because the third vector is the sum of the first two which makes it a linear combination of them

which means the first set is a spanning set but the second isnt

Example 3.7.9. consider the vector space $\mathbb{R}^4$, given the vectors $v_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ -6 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}, v =$

$\begin{bmatrix} 1 \\ 2 \\ 1 \\ -4 \end{bmatrix}$, check if $v \in \text{span}\{v_1, v_2, v_3\}$

this means there exist $x, y, z \in \mathbb{R}$ such that $x \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \\ -2 \end{bmatrix} + y \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \\ -6 \end{bmatrix} + z \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ -4 \end{bmatrix}$

to solve this we build an augmented matrix:

$$\begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \\ -1 & 3 & 1 \\ -2 & -6 & -1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ -4 \end{bmatrix} \implies \begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 2 & 0 & 2 \\ -1 & 3 & 1 & 1 \\ -2 & -6 & -1 & -4 \end{bmatrix}$$

we reduce to echelon form:

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 4 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

this matrix tells us that we have infinite x, y, z that give us what we want, therefore $v \in \text{span}\{v_1, v_2, v_3\}$

Example 3.7.10. $\mathbb{R}^n$ is spanned by the set $x_1, x_2 \dots x_n$:

$$\text{span}(\mathbb{R}^n) = \sum_{i=1}^{n} x_i u_i$$

$\mathbb{R}^n$ is spanned by the following n vectors:

$$\begin{aligned} u_1 &= (1, 0, 0 \dots 0) \\ u_2 &= (0, 1, 0 \dots 0) \\ u_2 &= (0, 0, 1 \dots 0) \\ &\vdots \\ u_n &= (0, 0, 0 \dots 1) \end{aligned}$$

Example 3.7.11. $\mathbb{R}[x]$ is the space of polynomials over the field $\mathbb{R}$ with a single polynomial x
$\mathbb{R}[x]$ isnt finitely spanned, to prove this, we assume in contradiction that the set $p_1(x), p_2(x) \dots p_n(x)$ spans $\mathbb{R}[x]$ a linear combination of polynomials cant give us a degree that is higher than the degree of the polynomial with the highest degree in said linear combination, we write:

$$\sum_{i=1}^{n} x_i p_i(x) \leq max\{deg(p_{1 \to n}(x))\}$$

let m be the polynomial with the highest degree, $m = max\{deg(p_{1 \to n}(x))\}$, we know $x^{m+1} \in \mathbb{R}[x]$ by necessity because $\mathbb{R}[x]$ contains all the polynomials over the field $\mathbb{R}$

$$\sum_{i=1}^{n} x_i p_i(x) \leq m \implies \sum_{i=1}^{n} x_i p_i(x) \neq x^{m+1} \implies x^{m+1} \notin \mathbb{R}[x] \implies \text{span}\{p_1 \dots p_n\} \neq \mathbb{R}[x]$$

Lemma 29. let V be a vector space over the field $\mathbb{F}$, and let $T \subseteq V, S \subseteq V$ be sets of vectors, then $\text{span}(S) \subseteq \text{span}(T) \iff S \subseteq \text{span}(T)$

Lemma 30. if $V_{\mathbb{F}}$ is finitely spanned, if $W \leq V$ then W is finitely spanned too

linear dependance

Definition 3.7.6. a set of vectors $v_1, v_2, \ldots, v_k$ from a vector space is said to be **linearly dependant** if there exists a non-trivial (not all zeros) set of scalars $a_1, a_2, \ldots, a_k$, such that:

$$a_1 v_1 + a_2 v_2 + \cdots + a_k v_k = \vec{0}$$

thus, a set of vectors is linearly dependent if and only if one of them is zero or a linear combination of the others.
each of the vectors in a linearly dependant set is linearly dependant on the set

> **note**
> a **set of vectors** is just a subset of a vector space V while a **sequence of vectors** is a map $\mathbb{N} \to V$ (can also be written as a infinite tuple). A set does not care about ordering or enumerating elements multiple times in contrast to a sequence.

Example 3.7.12. given $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$ check whether these vectors are linearly dependant

we need to find x, y, z such that $xv_1 + xv_2 + xv_3 = \vec{0}$, and the first combination that comes to mind is $v_1 + v_2 - v_3 = \vec{0}$, so these vectors are linearly dependant

Lemma 31. $\{\vec{v}\}$ is a linearly dependant set if and only if $\vec{v} = \vec{0}$

Proof. **first** we prove

$$\{\vec{v}\} \text{ being linearly dependant } \implies \vec{v} = \vec{0}$$

$\{\vec{v}\}$ being linearly dependant means there exists $a \neq 0 \in \mathbb{F}$ such that:

$$a\vec{v} = \vec{0}$$

we multiply both sides of this equation by $a^{-1} \in \mathbb{F}$

$$a^{-1}a\vec{v} = a^{-1}\vec{0}$$
$$\vec{v} = \vec{0}$$

second we prove

$$\vec{v} = \vec{0} \implies \{\vec{v}\} \text{ is linearly dependant}$$

we need to find $a \neq 0 \in \mathbb{F}$ such that

$$a\vec{v} = \vec{0}$$

and that a can just be $a = 1$ which would give us $1 \cdot \vec{v} = \vec{0}$, therefore $\{\vec{v}\}$ is linearly dependant

$\square$

proportionality

Definition 3.7.7. v_1, v_2 are called **proportional** if there exists $a \in \mathbb{F}$ such that $v_1 = a \cdot v_2$

Lemma 32. $\{v_1, v_2\}$ is a linearly dependant set if and only if the vectors v_1, v_2 are proportional

Proof. **first** we need to prove

$$\{v_1, v_2\} \text{ is linearly dependant} \implies \{v_1, v_2\} \text{ are proportional}$$

$\{v_1, v_2\}$ being linearly dependant means there exist $a, b \in \mathbb{F}$ not both 0 such that $av_1 + bv_2 = \vec{0}$
we assume **Without loss of generality** that $a \neq 0$
so we can manipulate the equation to be:

$$v_1 = -\frac{b}{a} v_2$$

and therefore v_1, v_2 are proportional
second we need to prove

$$v_1, v_2 \text{ are proportional} \implies \{v_1, v_2\} \text{ are linearly dependant}$$

v_1, v_2 being proportional means there exists $a \in \mathbb{F}$ such that:

$$v_1 = av_2$$

now we can manipulate this expression to be:

$$v_1 - av_2 = \vec{0}$$

which obviously tells us that $\{v_1, v_2\}$ are linearly dependant because there exists a set of coefficients not all 0 that gives us the vector $\vec{0}$

□

Lemma 33. if $S \subset T$ and both are finite sets, if S is linearly dependant then T is linearly dependant aswell

Proof. let $S = \{s_1, s_2, \ldots, s_n\}$, $T = \{s_1, s_2, \ldots, s_n, t_{n+1}, \ldots, t_{k+1}\}$
since S is linearly dependant then we have the scalars $x_1, x_2, \ldots, x_n \in \mathbb{F}$ not all zero that give us:

$$x_1 s_1 + x_2 s_2 + \cdots + x_n s_n = \vec{0}$$

consider the following sum:

$$x_1 s_1 + \cdots + x_n s_n + 0 \cdot t_{n+1} + \cdots + 0 \cdot t_{n+k} = \vec{0}$$

this sum shows just why T would be linearly dependant, because it is a linear combination whose scalars arent all 0 that gives us $\vec{0}$

□

Entailment 9. if a sequence of vectors $A = \{v_1, \ldots, v_n\}$ contains a zero vector $\vec{0}$ or 2 vectors that are proportional then A is linearly dependant

Lemma 34. let $A = \{v_1, \dots, v_n\}$ be a sequence of vectors such that $v_1 \neq \vec{0}$, therefore the following statements are equal

1. A is a linearly dependant set of vectors

2. the sequence A contains a vector $v_i,\ i \geq 2$ which is a linear combination of its preceding vectors

3. the sequence A contains a vector $v_i,\ i \geq 2$ that is a linear combination of other vectors which means $v_i \in \text{span}\{v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_n\}$

Proof. we start with proving statement 1 $\implies$ statement 2
according to statement 1, A is a linearly dependant set of vectors, therefore there exists a non-trivial linear combination that gives us $\vec{0}$

$$x_1v_1 + x_2v_2 + \cdots + x_nv_n = \vec{0}$$

we pick the last coefficient that isnt 0, let $x_k \neq 0$ be that coefficient, so we know $x_{k+1} \cdots x_n = 0$ because x_k is the last non-zero scalar
and we can also conclude that the following is a non-trivial sequence that gives us $\vec{0}$ because all scalars that come after x_k are 0

$$x_1v_1 + x_2v_2 + \cdots + x_kv_k = \vec{0}$$

we manipulate this expression:

$$\begin{aligned} x_1v_1 + x_2v_2 + \cdots + x_kv_k &= \vec{0} \\ x_1v_1 + x_2v_2 + \cdots + x_{k-1}v_{k-1} &= -x_kv_k \\ \frac{x_1}{-x_k}v_1 + \frac{x_2}{-x_k}v_2 + \cdots \frac{x_{k-1}}{-x_k}v_{k-1} &= v_k \end{aligned}$$

and so we have proved that v_k is a linear combination of its preceding vectors
not that $k \geq 2$ because if $k = 1$ and k is the rightmost non-zero scalar then all the scalars are zero

□

Proof. now we prove statement 2 $\implies$ statement 3
statement 2 tells us there exists $i \geq 2$ such that $v_i \in \text{span}\{v_1 \cdots v_{i-1}\}$ which means:

$$v_i = b_1v_1 + \cdots + b_{i-1}v_{i-1}$$

we can add the rest of the vectors with 0 as the coefficients and not affect the equation

$$v_i = b_1v_1 + \cdots + b_{i-1}v_{i-1} + 0 \cdot v_{i+1} + \cdots 0 \cdot v_n$$

□

Proof. now we prove statement 3 $\implies$ statement 1
there exist $x_1, x_2, x_{i-1}, x_{i+1}, x_n \in \mathbb{F}$ such that $v_i = x_1v_1 + x_{i-1}v_{i-1} + x_{i+1}v_{i+1} + x_nv_n$
we move v_i to the other side of the equation to get:

$$\vec{0} = x_1v_1 + x_{i-1}v_{i-1} + (-1)v_i + x_{i+1}v_{i+1} + x_nv_n$$

since $x_i = -1$, this shows that we have a non-trivial linear combination that gives us $\vec{0}$ which means that the set is linearly dependant

$\square$

Entailment 10. if $S \subseteq V$ is linearly dependant then there exists $\vec{s} \in S$ such that $\vec{s} \in \text{span}(S - \{s\})$
in other words, if a given sequence of vectors is linearly dependant then it has atleast one vector that is a linear combination of the other vectors

Lemma 35. let $S = \{s_1, s_2, \ldots, s_n\}$, if $S \subseteq V$ is a dependant set then there exists $\vec{s} \in S$ such that $\text{span}(s) = \text{span}(S - \{s\})$
in other words, given a dependant set of vectors, there exists a vector in this set that is possible to drop without changing the span of the set

Proof. according to this entailment, there exists $s \in S$ such that $s \in \text{span}(S - \{s\})$
to prove $\text{span}(s) = \text{span}(S - \{s\})$ we show that $\text{span}(s) \subseteq \text{span}(S - \{s\})$ and $\text{span}(S - \{s\}) \subseteq \text{span}(s)$
we know that $\text{span}(S - \{s\}) \subseteq \text{span}(s)$ because we can take every linear combination from the former, add $0 \cdot s$ to it, and it would be a linear combination of the latter
without loss of generality, assume $s = s_1$
since $s_1 \in \text{span}(S - \{s_1\})$ then:

$$s_1 = y_2 s_2 + y_3 s_3 + \cdots + y_n s_n$$

where $y_2 \ldots y_n \in \mathbb{F}$
let $v \in \text{span}(S)$ then:

$$v = x_1 s_1 + x_2 s_2 + \cdots + x_n s_n$$

where $x_2 \ldots x_n \in \mathbb{F}$
we substitute s_1 in this equation to get:

$$v = x_1(y_2 s_2 + y_3 s_3 + \cdots + y_n s_n) + x_2 s_2 + \cdots + x_n s_n$$
$$v = (x_1 y_2 + x_2) s_2 + \cdots + (x_1 y_n + x_n) s_n$$

we have showed that $v \in \text{span}(S - \{s_1\})$
and since $v \in \text{span}(S)$ and $v \in \text{span}(S - \{s\})$ then $\text{span}(S) \subseteq \text{span}(S - \{s\})$
and since $\text{span}(s) \subseteq \text{span}(S - \{s\})$ and $\text{span}(S - \{s\}) \subseteq \text{span}(s)$ then $\text{span}(s) = \text{span}(S - \{s\})$

$\square$

Lemma 36. for every finite set $S \in V_{\mathbb{F}}$, there exists $B \subseteq S$ such that:

1. B is linearly indepenedant

2. $\text{span}(B) = \text{span}(S)$

Proof. in case S is linearly independant we pick $B = S$ and so B is linearly dependant and $\text{span}(B) = \text{span}(S)$
in case S is linearly dependant we got more work to do:
according to this lemma, $\text{span}(S) = \text{span}(S - \{s_1\})$
in case $S - \{s_1\}$ is linearly independant then let $B = S - \{s_1\}$ which means $B \subseteq S$ and $\text{span}(B) = \text{span}(S)$
in case $S - \{s_1\}$ is linearly dependant, according to this lemma, $\text{span}(S - \{s_1, s_2\}) = \text{span}(S - \{s_1\}) = \text{span}(S)$

in case $\text{span}(S - \{s_1, s_2\})$ is linearly independant then the proof is over
in case it isnt, we repeat the same step of removing a vector
you might already see the pattern here, we keep dropping vectors from S without changing the span, eventually we would arrive at $S - \{s_1, s_2, s_k\}$ that is linearly independant and $B \subseteq S$ and $\text{span}(B) = \text{span}(S)$

□

3.7.4 basis

Definition 3.7.8. assuming a vector space V, let $B \subseteq V$ be a finite, linearly independant and spanning set of V which we call a **basis**, each vector in a basis is called a **basis vector**
let $B = \{u_1, \dots, u_i, \dots, u_n\}$ be a basis of $\mathbb{R}^n$:

$$\begin{aligned} u_1 &= (1, 0, \dots, 0) \\ u_2 &= (0, 1, \dots, 0) \\ &\vdots \\ u_i &= (0, 0, \dots, 1 \dots, 0) \\ &\vdots \\ u_n &= (0, 0, \dots, 1) \end{aligned}$$

since B is linearly independant then:

$$\sum_{i=1}^{n} x_i u_i = \vec{0} \iff x_1 = x_2 = \dots = x_n = 0$$

this is necessarily true because x_i has to be 0 for u_i to zero out and if it wasnt zero then the output vector would have a non-zero in it and it wouldnt be $\vec{0}$ anymore

> **note**
> a basis doesnt have to be of this simplified form, we can take the basis B, apply elementary row operations to its vectors and we would get another basis for the given vector space

Lemma 37. every **finitely generated vector space** $V_{\mathbb{F}}$ has atleast one basis

Proof. since $V_{\mathbb{F}}$ is finitely generated there exists a finite set $S \subseteq V$ such that $\text{span}(S) = V$
according to this lemma, there exists $B \subseteq S$ where B is linearly independant and $\text{span}(B) = V$, therefore B is a basis

□

Example 3.7.13. $V = \{\vec{0}\}$
$\text{span}(\varnothing) = \{\vec{0}\}$
$\varnothing$ is linearly independant
$\varnothing$ is a basis of V

Example 3.7.14. given the vector space $W = \{(x, y, z) \in \mathbb{R}^3 \mid x = y\}$, we find a basis of it:

$$(x, y, z) = (x, x, z) = x(1, 1, 0) + z(0, 0, 1) \implies B = \{(1, 1, 0), (0, 0, 1)\}$$

so B is a basis of W and $\dim W = 2$, see dimension

Example 3.7.15. consider the vector space $M_{n\times m}(\mathbb{F})$ which represents all the matrices with dimensions $n \times m$ over the field $\mathbb{F}$
what we are basically looking for is a set of matrices that is linearly independant so that no matrix is a linear combination of the others, which is accomplished with:

$$u_{i,j} = i \begin{array}{c} \\ \\ \end{array} \begin{array}{c} \\ \\ \end{array} \begin{array}{c} j \\ \begin{pmatrix} 0 & \vdots & 0 \\ \cdots & 1 & \\ 0 & & 0 \end{pmatrix} \end{array}$$

and:

$$\underbrace{\text{span}\{u_{i,j}\}}_{n\cdot m \text{ elements}} = M_{n\times m}(\mathbb{F})$$

we prove that $\{u_{i,j}\}$ is linearly independant:

$$\sum_{i=1}^{n}\sum_{j=1}^{m} \alpha_{i,j} \cdot u_{i,j} = \vec{0}$$

it must be that $a_{i,j} = 0$ and so $\{u_{i,j}\}$ is linearly independant because the only linear combination that gives us $\vec{0}$ is where the coefficients all are 0

Lemma 38. given S is a finite set and $T \subseteq \text{span}(S)$ and $|T| > |S|$, then T is linearly dependant
in other words, if we have a finitely generated vector space, if T is a subset of that space such that the number of elements in T is bigger than the number of elements in S, then T is linearly dependant

Proof. let $S = \{s_1, s_2 \dots s_n\}$ and $T = \{t_1, t_2 \dots t_n\}$ such that $n > m$
let A denote the coefficients, for every $1 \leq i \leq n$ where $t_i \in \text{span}(S)$ there exist the coefficients:

$$t_i = \sum_{j=1}^{m} A_{j,i} S_j$$

□

Entailment 11. let S be a finite set, if $T \subseteq \text{span}(S)$ and T is linearly independant then $|T| \leq |S|$

Entailment 12. let A, B be bases of V then $|A| = |B|$

Proof. B is a spanning set and $A \subseteq \text{span}(B)$ and A is linearly independant then $|A| \leq |B|$
A is a spanning set and $B \subseteq \text{span}(A)$ and B is linearly independant then $|B| \leq |A|$
therefore $|B| = |A|$

□

standard basis

Definition 3.7.9. the **standard basis** of a vector space is the set of vectors whose components are all zero, except one that equals 1.

Example 3.7.16. consider the set $\{(1,0),(0,1)\}$ which is a standard basis of the vector space $\mathbb{R}^2$
also the dimension of this set is 2

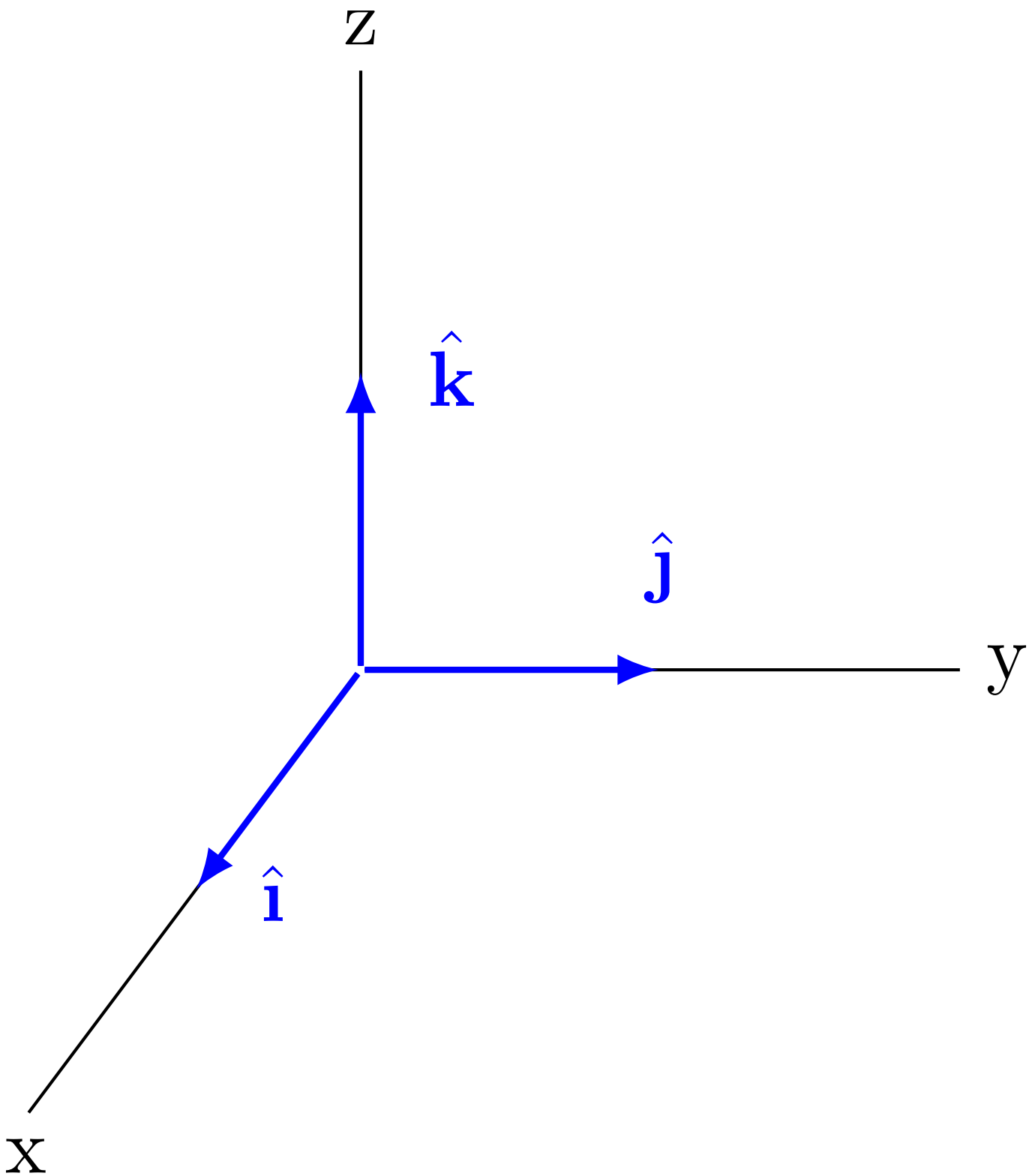
Lemma 39. when put in a matrix, a standard basis forms an identity matrix

Example 3.7.17. consider the set $A = \{(1,0,0),(0,1,0),(0,0,1)\}$ which is a standard basis of the vector space $\mathbb{R}^3$
we put the vectors in a matrix:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

which is an identity matrix of the 3rd degree
$\dim A = 3$

basis of the cartesian coordinate system the standard basis of the cartesian coordinate system consists of the unit vectors that lie along the x, y, and z axes
the x unit vector is denoted by $\hat{\mathbf{i}}$, the y unit vector by $\hat{\mathbf{j}}$, and the z unit vector by $\hat{\mathbf{k}}$



Characteristic 18.

$$\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 1$$
$$\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{k}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{i}} = 0$$
$$\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}$$
$$\hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}$$
$$\hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$$

we can write any vector in terms of the base vectors
$A = A_x\hat{\mathbf{i}} + A_y\hat{\mathbf{j}} + A_z\hat{\mathbf{k}}$

3.7.5 dimension

Definition 3.7.10. the **dimension** of a vector space $V_{\mathbb{F}}$ is the number of elements in one of its bases
we write $\dim V_{\mathbb{F}} = |B|$ where B is a basis of V

Example 3.7.18. $\dim(\mathbb{F}^n) = n$

Example 3.7.19. $\dim(M_{n\times m}(\mathbb{F})) = n \times m$

Example 3.7.20. $\dim(P_n[x]) = n + 1$

Example 3.7.21. $\dim(\ker(A)) =$ number of free variables

Lemma 40. let $V_{\mathbb{F}}$ be a vector space of dimension n, for every subset $S \subseteq V$ that has n elements the following statements are equal

1. S is linearly independant

2. $\text{span}(S) = V$

Proof.

Step 2.0.1.1. we prove statement 1 $\implies$ statement 2
given $S = \{s_1, s_2 \dots s_n\} \subseteq V^n$ is linearly independant, we need to prove $\text{span}(S) = V$
let $v \in V$, B a basis of V, $|B| = n$
let $T = \{s_1, s_2 \dots s_n, v\}$
we know according to previous lemmas that $|T| > n$ therefore T is linearly dependant
since T is linearly dependant then we have the following non-trivial linear combination that gives us $\vec{0}$:

$$\sum_{i=1}^{n} x_i s_i + s_{n+1} v = \vec{0}$$

we know $x_{n+1} \neq 0$ because otherwise another s_i would be equal to 0 and S wouldnt be linearly independant
by manipulating the expression we get:

$$\sum_{i=1}^{n} \frac{x_i}{x_{n+1}} \cdot s_i = v$$

therefore $v \in \text{span}(S)$ therefore $\text{span}(S) = V$

Step 2.0.1.2. we prove statement 2 $\implies$ statement 1
given $\text{span}(S) = V^n$
assume in contradiction that S is linearly dependant
according to a previous lemma we there exists $B \subset S$ that is linearly independant such that $\text{span}(B) = \text{span}(S)$
therefore B is a basis therefore $|B| = n$
we arrived at a contradiction becacuse $|B| < |S| = n$

□

Entailment 13. if $\dim V = m$ and $B = \{u_1, u_2 \dots u_m\}$, if $\text{span}(B) = V$ or B is linearly independant then B is a basis of $V_{\mathbb{F}}$

Lemma 41. given $V_{\mathbb{F}}$ is finitely spanned, if $W \leq V$ then $\dim W \leq \dim V$

> **note**
> according to the previous lemma, since V is finitely spanned then W is finitely spanned too which means W has a basis and a dimension

we assume in contradiction that $\dim W > \dim V$
assume $\{v_1 \dots v_n\}$ is a basis of V, then we know $\text{span}\{v_1 \dots v_n\} = V$
assume $\{w_1 \dots w_m\}$ where $m > n$ is a basis of W, then we know $\text{span}\{w_1 \dots w_m\} = W$ which means $\{w_1 \dots w_m\}$ is linearly independant but it cant be because we have a set where $m \leq n$ with less elements that is linearly independant so a set with $m > n$ cant be linearly independant

Lemma 42. given ${V_{\mathbb{F}}}^n$ is finitely spanned, ${W}^n \leq {V}^n$ and $\dim W = \dim V$ then $W = V$

Proof. let $w_1 \dots w_n$ be a basis of W and so $n = |V|$ and $w_1 \dots w_n$ is linearly independant therefore according to this lemma $\text{span}\{w_1 \dots w_n\} = V = W$ and since it spans V and is of size $n = |V|$ then it is a basis of V

□

Lemma 43. if $V_{\mathbb{F}}$ is finitely spanned and $W \leq V$ then for every basis B of W there exists a basis A of V such that $B \subseteq A$
we call this operation as complementing the basis B to the basis A

Proof. we prove using induction of $\dim V - \dim W$

Step 2.0.1.3. the first step of an induction proof is a check
we check if $\dim V - \dim W = 0$ meaning $\dim V = \dim W$
we pick $A = B$

to be continued...

□

Lemma 44. let V be a finitely spanned vector space, let $U, W \leq V$ then:

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$$

Proof. with previous lemmas in mind, we know $U+W,\ U\cap W \leq V$
let $a_1 \dots a_n$ be a basis of $U+W$
since $U\cap W \leq V$ then we can complement $a_1 \dots a_n$ to a basis of U such that $a_1 \dots a_n + a_{n+1} \dots a_m$ is a basis of U
and since $U \cap W \leq W$ then we can complement $a_1 \dots a_n$ to a basis of W such that $a_1 \dots a_n + b_{n+1} \dots b_k$ is a basis of W
so far we know $\dim W = k,\ \dim U = m,\ \dim(W\cap U) = n$
to be continued...

□

3.7.6 coordinate vector

Definition 3.7.11. let V_F be a vector space such that $\dim V = m$, and $B = \{v_1, v_2 \dots v_m\}$ be a basis of V and $u \in V$
the **coordinate vector** of u relative to the basis B is defined as $[u]_B = (k_1, k_2 \dots k_m)$ such that $k_1v_1 + k_2v_2 + \dots + k_mv_m = u$

Question 3. given the vector $(18, 21, 20)$ and the basis $(1,2,3), (4,5,6), (7,8,8)$ of $\mathbb{R}^3$, find the coordinates of the given vector relative to the given basis

Answer 3.0.1. we place these vectors in a matrix and reduce it:

$$\left(\begin{array}{ccc|c} 1 & 4 & 7 & 18 \\ 2 & 5 & 8 & 21 \\ 3 & 6 & 8 & 20 \end{array}\right) \xrightarrow[r_3-3r_1\to r_3]{r_2-2r_1\to r_2} \left(\begin{array}{ccc|c} 1 & 4 & 7 & 18 \\ 0 & -3 & -6 & -15 \\ 0 & -6 & -13 & -34 \end{array}\right)$$
$$\xrightarrow{\frac{r_2}{-3}\to r_2} \left(\begin{array}{ccc|c} 1 & 4 & 7 & 18 \\ 0 & 1 & 2 & 5 \\ 0 & -6 & -13 & -34 \end{array}\right)$$
$$\xrightarrow[r_3+6r_2\to r_3]{r_1-4r_2\to r_1} \left(\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & -1 & -4 \end{array}\right)$$
$$\xrightarrow{\frac{r_3}{-1}\to r_3} \left(\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 1 & 4 \end{array}\right)$$
$$\xrightarrow[r_2-2r_3\to r_2]{r_1+r_3\to r_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 4 \end{array}\right)$$

which means $[(18, 21, 20)]_B = (2, -3, 4)$

transformation matrix

Definition 3.7.12. let V_F be a vector space such that $\dim v = m$ and let $A = \{u_1, u_2 \dots u_m\}, B = \{v_1, v_2 \dots v_m\}$ be bases of V, the **transformation matrix** from B to A is defined as ${}_BT_A = ([u_1]_B^T, [u_2]_B^T, \dots, [u_m]_B^T)$

Question 4. consider the following bases of $\mathbb{R}^3$:

$$u = \{(1,0,0), (1,1,0), (1,1,1)\}$$
$$v = \{(1,1,1), (0,1,1), (0,0,1)\}$$

find the following:

1. $[u_1]_v$

2. $[u_2]_v$

3. $[u_3]_v$

4. ${}_vT_u$

instead of putting each of the first 3 in its own matrix and reducing we use 1 matrix for them all

$$\left(\begin{array}{ccc|ccc} v_1 & v_2 & v_3 & u_1 & u_2 & u_3 \\ \hline 1 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow[r_3-r_1\to r_3]{r_2-r_1\to r_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & -1 & -1 & 0 \end{array}\right)$$

$$\xrightarrow{r_3-r_2\to r_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 \end{array}\right)$$

which gives us:

1. $[u_1]_v = [(1,0,0)]_v = (1,-1,0)$

2. $[u_2]_v = [(1,1,0)]_v = (1,0,-1)$

3. $[u_3]_v = [(1,1,1)]_v = (1,0,0)$

4. ${}_vT_u = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix}$

Lemma 45. let V_F be a vector space such that $\dim V = m$ and let $A = \{u_1, u_2 \dots u_m\}, B = \{v_1, v_2 \dots v_m\}$ be bases of V then:

1. $A \cdot_A T_B = B$

2. ${}_BT_A = ({}_AT_B)^{-1}$

3.7.7 vector space addition

Definition 3.7.13. given the vector spaces U, W that are subspaces of the vector space $V_{\mathbb{F}}$ (meaning $U, W \leq V$), then the addition of U and W is defined as:

$$U + W = \{u + w \mid u \in U \wedge w \in W\}$$

Lemma 46.

$$U + W = \{u + w \mid u \in U \wedge w \in W\} = \text{span}(\text{basis of } U \cup \text{basis of } W)$$

Lemma 47. $U + W \leq V$

Proof.

$$\begin{aligned} U + W \subseteq V \implies & u \in U \subseteq V \\ & w \in W \subseteq V \\ & u + w \in V && \text{addition closure under } V \end{aligned}$$

Step 4.0.0.1. step 1: prove $\vec{0} \in U + W$

$$\begin{aligned} \vec{0} &\in U && \text{because } U \text{ is a subspace} \\ \vec{0} &\in W && \text{because } W \text{ is a subspace} \\ \vec{0} + \vec{0} = \vec{0} &\in W + U \end{aligned}$$

Step 4.0.0.2. step 2: prove addition closure
let $x_1, x_2 \in \mathbb{U} + W$

$$\begin{aligned} u_1 + w_1 &= x_1 \\ u_2 + w_2 &= x_2 \\ x_1 + x_2 = (u_1 + w_1) + (u_2 + w_2) &= (u_1 + u_2) + (w_1 + w_2) \in W + U \end{aligned}$$

Step 4.0.0.3. step 3: prove multiplication closure
let $x \in U + W,\ a \in \mathbb{F},\ u_1 + w_1 = x$

$$ax = a(u_1 + w_1) = au_1 + aw_1 \in U + W$$

□

Example 3.7.22. given $U = \text{span}(\{(1,2,1,2,1),(2,1,2,1,2)\}), V = \text{span}(\{(0,1,2,3,0),(3,2,1,0,3)\})$, we find the basis and dimension of $U + V$:
$U + V = \text{span}(\{(1,2,1,2,1),(2,1,2,1,2),(0,1,2,3,0),(3,2,1,0,3)\})$
we put these vectors in a matrix and reduce it to find out how many linearly independant vectors we have

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 1 & 1 & 2 \\ 1 & 2 & 2 & 1 \\ 2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 3 \end{bmatrix} \xrightarrow{\text{row reduction}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

which means that only the first 3 vectors are linearly independant so $U+V = \text{span}(\{(1,2,1,2,1),(2,1,2,1,2),(0,1,2$
and $\dim(U + V) = 3$

3.7.8 vector space intersection

Definition 3.7.14. given the vector spaces U, W that are subspaces of the vector space $V_{\mathbb{F}}$ (meaning $U, W \leq V$), then the intersection of U and W is defined as:

$$U \cap W = \{v \mid v \in U \wedge v \in W\}$$

Lemma 48. $U \cap W \leq V$

Proof.

$$\begin{aligned}
U \cap W &\subseteq V && \text{because they are subspaces so they must be subsets} \\
\vec{0} &\in U && \text{because } U \leq V \\
\vec{0} &\in W && \text{because } W \leq V \\
\implies \vec{0} &\in U \cap W \\
x_1,\ x_2 &\in U \cap W && \text{assumption} \\
\implies x_1 + x_2 &\in U \cap W && \text{because both are subspaces of } V \\
a \in \mathbb{F},\ x &\in U \cap W && \text{assumption} \\
\implies a \cdot x &\in U \cap W && \text{because both are subspaces of } V
\end{aligned}$$

□

Example 3.7.23. given $U = \text{span}(\{(1,2,1,2,1),(2,1,2,1,2),(3,3,3,3,3)\}), V = \text{span}(\{(0,1,2,3,0),(3,2,1,0,3)\})$ we find the basis and dimension of $U \cap V$:
to find $U \cap V$ we need to write U and V as a system of equations

Step 4.0.0.4. to find the corresponding equations of U we reduce the following matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & x_1 \\ 2 & 1 & 3 & x_2 \\ 1 & 2 & 3 & x_3 \\ 2 & 1 & 3 & x_4 \\ 1 & 2 & 3 & x_5 \end{array}\right] \xrightarrow{\text{row reduction}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & x_1 \\ 0 & -3 & -6 & x_2 - 2x_1 \\ 0 & 0 & 0 & x_3 - x_1 \\ 0 & 0 & 0 & x_4 - x_2 \\ 0 & 0 & 0 & x_5 - x_1 \end{array}\right]$$

and so $U = \text{span}(\{(1,2,1,2,1),(2,1,2,1,2)\}),\ \dim U = 2$, and:

$$U = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \middle| \begin{array}{c} x_3 - x_1 = 0 \\ x_4 - x_2 = 0 \\ x_5 - x_1 = 0 \end{array} \right\}$$

Step 4.0.0.5. now to find the corresponding equations of V we reduce the following matrix:

$$\left[\begin{array}{cc|c} 3 & 0 & x_1 \\ 2 & 1 & x_2 \\ 1 & 2 & x_3 \\ 0 & 3 & x_4 \\ 3 & 0 & x_5 \end{array}\right] \xrightarrow{\text{row reduction}} \left[\begin{array}{cc|c} 1 & 2 & x_3 \\ 0 & -3 & x_2 - 2x_3 \\ 0 & 0 & x_1 - 2x_2 + x_3 \\ 0 & 0 & x_2 - 2x_3 + x_4 \\ 0 & 0 & -2x_2 + x_3 + x_5 \end{array}\right]$$

and so $V = \text{span}(\{(3,2,1,0,3),(0,1,2,3,0)\}),\ \dim V = 2$, and:

$$V = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \middle| \begin{array}{c} x_1 - 2x_2 + x_3 = 0 \\ x_2 - 2x_3 + x_4 = 0 \\ -2x_2 + x_3 + x_5 = 0 \end{array} \right\}$$

Step 4.0.0.6.

$$U \cap V = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \middle| \begin{aligned} x_3 - x_1 &= 0 \\ x_4 - x_2 &= 0 \\ x_5 - x_1 &= 0 \\ x_1 - 2x_2 + x_3 &= 0 \\ x_2 - 2x_3 + x_4 &= 0 \\ -2x_2 + x_3 + x_5 &= 0 \end{aligned} \right\}$$

we put these equations in a matrix and reduce it:

$$\left(\begin{array}{ccccc|c} -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 \\ 0 & -2 & 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 \end{array}\right) \xrightarrow{\text{row reduction}} \left(\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

which gives us:

$$\begin{aligned} x_1 &= x_5 \\ x_2 &= x_5 \\ x_3 &= x_5 \\ x_4 &= x_5 \\ (x_1, x_2, x_3, x_4, x_5) &= (x_5, x_5, x_5, x_5, x_5) = x_5(1, 1, 1, 1, 1) \end{aligned}$$

and so $U \cap V = \text{span}(\{(1, 1, 1, 1, 1)\})$ and $\dim(U \cap V) = 1$

3.7.9 vector space union

4 linear map

Definition 4.0.1. let U, V be vector spaces over the field $\mathbb{F}$, and let $L : U \to V$ be a function which would be called a **linear map** if:

1. for every $u_1, u_2 \in U$, $L(u_1 + u_2) = L(u_1) + L(u_2)$

2. for every $u \in U$ and $k \in \mathbb{F}$, $L(ku) = kL(u)$

Question 5.

$$L : \mathbb{R}^3 \to \mathbb{R}^3$$
$$L(x, y, z) = \begin{pmatrix} x - y - z \\ 2x - y + 2z \\ 3x - 2y + z \end{pmatrix}$$

is L a linear map?

Answer 5.0.1.

Step 5.0.1.1. we need to prove $L((x_1, y_1, z_1) + (x_2, y_2, z_2)) = L(x_1, y_1, z_1) + L(x_2, y_2, z_2)$

note
i should be writing $L((x,y,z))$ not $L(x,y,z)$

$$\begin{aligned}
L((x_1,y_1,z_1)+(x_2,y_2,z_2)) &= L(x_1+x_2, y_1+y_2, z_1+z_2) \\
&= \begin{pmatrix} x_1+x_2-(y_1+y_2)-(z_1+z_2) \\ 2(x_1+x_2)-(y_1+y_2)+2(z_1+z_2) \\ 3(x_1+x_2)-2(y_1+y_2)+(z_1+z_2) \end{pmatrix} \\
&= \begin{pmatrix} (x_1-y_1-z_1)+(x_2-y_2-z_2) \\ (2x_1-y_1+2z_1)+(2x_2-y_2+2z_2) \\ (3x_1-2y_1+z_1)+(3x_2-2y_2+z_2) \end{pmatrix} \\
&= \begin{pmatrix} x_1-y_1-z_1 \\ 2x_1-y_1+2z_1 \\ 3x_1-2y_1+z_1 \end{pmatrix} + \begin{pmatrix} x_2-y_2-z_2 \\ 2x_2-y_2+2z_2 \\ 3x_2-2y_2+z_2 \end{pmatrix} \\
&= L(x_1,y_1,z_1)+L(x_2,y_2,z_2)
\end{aligned}$$

Step 5.0.1.2. we need to prove $L(k(x,y,z)) = kL(x,y,z)$

$$\begin{aligned}
L(k(x,y,z)) &= L(kx,ky,kz) \\
&= \begin{pmatrix} kx-ky-kz \\ 2kx-ky+2kz \\ 3kx-2ky+kz \end{pmatrix} \\
&= k\begin{pmatrix} x-y-z \\ 2x-y+2z \\ 3x-2y+z \end{pmatrix} \\
&= kL(x,y,z)
\end{aligned}$$

Question 6. check whether this represents a linear map:

$$T : \mathbb{R}^3 \to \mathbb{R}^3 := T(x,y,z) = (x^2, x+y, y+z)$$

Answer 6.0.1. it isnt, a counter example:

$$\begin{gathered}
T(1,0,0) = (1,1,0) \\
T(2,0,0) = (4,2,0) \\
T(1,0,0)+T(2,0,0) = (1,1,0)+(4,2,0) = (5,3,0) \\
T((1,0,0)+(2,0,0)) = T(3,0,0) = (9,3,0) \neq (5,3,0)
\end{gathered}$$

Lemma 49. if $L : U \to V$ is a linear map then $L(\vec{0}) = \vec{0}$

Entailment 14. if $f : U \to V$ is a function such that $f(\vec{0}) \neq \vec{0}$ then f isnt a linear map

Lemma 50. if $L : U \to V$ is a linear map then:

1. the kernel of L is $\ker L = \left\{ u \in U \mid L(u) = \vec{0} \right\}$

2. the **image** of L is $\operatorname{Im} L = \{L(u) \mid u \in U\} \leq V$, which means all the values that can be returned by L, which is also the column space of its corresponding transformation matrix (if it has one)

Question 7. given the linear operator $L : \mathbb{R}^4 \to \mathbb{R}^4$ defined by $L((x, y, z, w)) = (x + y, y + z, z + w, w + x)$

Subquestion 7.1. find the matrix of L in the basis $b_1 = (1, 1, 1, 1), b_2 = (1, -1, 0, 0), b_3 = (0, 1, -1, 0), b_4 = (0, 0, 1, -1)$

Answer 7.1.1.

$$L(\overset{x}{1}, \overset{y}{1}, \overset{z}{1}, \overset{w}{1}) = (1 + 1,\ 1 + 1,\ 1 + 1,\ 1 + 1) = (2, 2, 2, 2, 2)$$
$$L(\overset{x}{1}, \overset{y}{-1}, \overset{z}{0}, \overset{w}{0}) = (1 - 1,\ -1 + 0,\ 0 + 0,\ 0 + 1) = (0, -1, 0, 1)$$
$$L(0, 1, -1, 0) = (1, 0, -1, 0)$$
$$L(0, 0, 1, -1) = (0, 1, 0, -1)$$

on the left columns of the matrix lies the basis and on the right 4 columns lies the image

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 2 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 & 2 & -1 & 0 & 1 \\ 1 & 0 & -1 & 1 & 2 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1 & 2 & 1 & 0 & -1 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 \end{bmatrix}$$

and so the matrix is:

$$[L]_A^B = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \end{bmatrix}$$

Subquestion 7.2. find the bases of $\ker L, \operatorname{Im} L$

Answer 7.2.1.

Step 7.2.1.1. finding the basis of $\ker L$

$$L(x,y,z,w) = (0,0,0,0)$$
$$(x+y,\ y+z,\ z+w,\ w+x) = (0,0,0,0)$$
$$x+y=0$$
$$y+z=0$$
$$z+w=0$$
$$x+w=0$$
$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{array}\right) \Longrightarrow \left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$
$$w \text{ free variable}$$
$$x = -w$$
$$y = w$$
$$z = -w$$
$$(x,y,z,w) = (-w,w,-w,w) = w(-1,1,-1,1)$$
$$\text{basis of } \ker L = \{(-1,1,-1,1)\}$$
$$\dim \ker L = 1$$

Step 7.2.1.2. to find the basis of $\operatorname{Im} L$ we reduce the matrix:

$$\begin{pmatrix} 2 & 0 & 1 & 0 \\ 2 & -1 & 0 & 1 \\ 2 & 0 & -1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix} \Longrightarrow \begin{pmatrix} 2 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

and so the basis of $\operatorname{Im} L$ is $B = \{(2,2,2,2),(0,-1,0,1),(1,0,-1,0)\}$ and $\dim \operatorname{Im} L = 3$
we apply a simple check: $\dim \mathbb{R}^4 = \dim \ker L + \dim \operatorname{Im} L = 1 + 3 = 4$ ✓

Question 8. given the linear operator $L : M_2(\mathbb{R}) \to M_2(\mathbb{R})$ defined by $L(X) = XA + AX$ such that $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

Subquestion 8.1. find the matrix of L in the basis $E_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, E_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, E_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, E_4 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

Answer 8.1.1.

$$
\begin{aligned}
L\begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix} &= \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix} \\
&= \begin{pmatrix} x_1+x_2 & x_1+x_2 \\ x_3+x_4 & x_3+x_4 \end{pmatrix} + \begin{pmatrix} x_1+x_3 & x_2+x_4 \\ x_1+x_3 & x_2+x_4 \end{pmatrix} \\
&= \begin{pmatrix} 2x_1+x_2+x_3 & x_1+2x_2+x_4 \\ x_1+2x_3+x_4 & x_2+x_3+2x_4 \end{pmatrix}
\end{aligned}
$$

$$L\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

$$L\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

$$L\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$$

$$L\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$$

$$\begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix} \sim \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \implies \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 2 & 2 & 2 & 0 \\ 0 & 1 & 0 & 1 & 2 & 2 & 0 & 2 \\ 0 & 1 & 0 & -1 & 2 & 2 & 0 & -2 \\ 1 & 0 & -1 & 0 & 2 & 2 & -2 & 0 \end{array}\right) \implies \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 2 & 2 & -2 & 0 \\ 0 & 1 & 0 & 0 & 2 & 2 & 0 & -2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 \end{array}\right)$$

$$[L] = \begin{pmatrix} 2 & 2 & -2 & 0 \\ 2 & 2 & 0 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

Subquestion 8.2. find the bases of $\ker L, \operatorname{Im} L$

Question 9. given the function $L : \mathbb{R}_2[x] \to M_2(\mathbb{R})$ defined by $L(p(x)) = \begin{pmatrix} p(1) & p(0) \\ p(0) & -p(1) \end{pmatrix}$

Subquestion 9.1. prove L is a linear transformation

Answer 9.1.1. L's domain is a vector space of polynomials of the second degree at most, $\mathbb{R}_2[x] = \{a_2x^2 + a_1x + a_0 \mid a_2, a_1, a_0 \in \mathbb{R}\}$

Step 9.1.1.1. assume $p_1(x), p_2(x) \in \mathbb{R}_2[x]$, need to prove $L((p_1+p_2)(x)) = L(p_1(x)) + L(p_2(x))$

$$
\begin{aligned}
L((p_1+p_2)(x)) &= \begin{pmatrix} (p_1+p_2)(1) & (p_1+p_2)(0) \\ (p_1+p_2)(0) & -((p_1+p_2)(1)) \end{pmatrix} \\
&= \begin{pmatrix} p_1(1)+p_2(1) & p_1(0)+p_2(0) \\ p_1(0)+p_2(0) & -p_1(1)-p_2(1) \end{pmatrix} \\
&= \begin{pmatrix} p_1(1) & p_1(0) \\ p_1(0) & -p_1(1) \end{pmatrix} + \begin{pmatrix} p_2(1) & p_2(0) \\ p_2(0) & -p_2(1) \end{pmatrix} \\
&= L(p_1(x)) + L(p_2(x))
\end{aligned}
$$

Step 9.1.1.2. let $p(x) \in \mathbb{R}_2[x],\ k \in \mathbb{R}$, need to prove $L(kp(x) = kL(p(x))$

$$\begin{aligned} L((kp)(x)) &= \begin{pmatrix} (kp)(1) & (kp)(0) \\ (kp(0)) & (-kp)(1) \end{pmatrix} \\ &= \begin{pmatrix} kp(1) & kp(0) \\ kp(0) & -kp(1) \end{pmatrix} \\ &= k \begin{pmatrix} p(1) & p(0) \\ p(0) & -p(1) \end{pmatrix} \\ &= kL(p(x)) \end{aligned}$$

therefore, L indeed is a linear transformation

Subquestion 9.2. find the matrix of L in standard basis form

Subquestion 9.3. find the bases of $\operatorname{Im} L, \ker L$

4.1 eigenvalues and eigenvectors

Definition 4.1.1. an **eigenvector** of a linear transformation is a nonzero vector that changes at most by a scalar factor when that linear transformation is applied to it. The corresponding **eigenvalue**, often denoted by λ, is the factor by which the eigenvector is scaled.

$$Av = \lambda v, v \neq 0 \in \mathbb{R}^n$$

here A is a transformation matrix, v is an eigenvector, and λ is the eigenvalue corresponding to v

4.1.1 characteristic polynomial

Definition 4.1.2. considering the formula $Av = \lambda v, v \neq 0 \in \mathbb{R}^n$ from the definition of eigenvalues and eigenvectors:

$$Av = \lambda v \implies Av - \lambda v = \vec{0} \implies Av - \lambda I v = \vec{0} \implies (A - \lambda I)v = \vec{0} \wedge v \neq \vec{0} \implies \infty \text{ solutions} \implies |A - \lambda I| = 0$$

we call $p(\lambda) = |A - \lambda I|$ the **characteristic polynomial** of A

4.1.2 eigenspace

Definition 4.1.3. an **eigenspace** is the set of eigenvectors corresponding to the same eigenvalue, plus the zero vector. the basis of an eigenspace is called an **eigenbasis**

Question 10. given the matrix

$$A = \begin{pmatrix} 1 & 1 & -1 \\ -1 & 3 & -1 \\ -1 & 1 & 1 \end{pmatrix}$$

Subquestion 10.1. find the characteristic polynomial and the eigenvalues,eigenvectors, and eigenspaces of A

Answer 10.1.1.

Step 10.1.1.1.

$$0 = |A - \lambda I| = \left| \begin{pmatrix} 1 & 1 & -1 \\ -1 & 3 & -1 \\ -1 & 1 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = \left| \begin{pmatrix} 1-\lambda & 1 & -1 \\ -1 & 3-\lambda & -1 \\ -1 & 1 & 1-\lambda \end{pmatrix} \right|$$

$$(1-\lambda)^2(3-\lambda) + 1 + 1 - (-1(1-\lambda) - 1(1-\lambda) + (3-\lambda)) = 0$$

$$p(\lambda) = -\lambda^3 + 5\lambda^2 - 8\lambda + 4$$

Step 10.1.1.2. to find the eigenvalues first we need to find the roots of the characteristic polynomial

$$-\lambda^3 + 5\lambda^2 - 8\lambda + 4 = 0$$

$$\lambda = 1, 2$$

$$A - I\lambda = \begin{pmatrix} 1-\lambda & 1 & -1 \\ -1 & 3-\lambda & -1 \\ -1 & 1 & 1-\lambda \end{pmatrix}$$

we substitute the eigenvalue $\lambda = 2$

$$\begin{aligned} \lambda = 2 &\implies \begin{pmatrix} 1-2 & 1 & -1 \\ -1 & 3-2 & -1 \\ -1 & 1 & 1-2 \end{pmatrix} \\ &\implies \begin{pmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &\implies x_1 = x_2 - x_3 \\ &\implies (x_1, x_2, x_3) = (x_2 - x_3, x_2, x_3) = x_2(1,1,0) + x_3(-1,0,1) \end{aligned}$$

the eigenvectors for the eigenvalue $\lambda = 2$ are $(-1,0,1), (1,1,0)$, the eigenspace for $\lambda = 2$ is $\text{span}\{(1,1,0), (-1,0,1)\}$ of dimension 2
therefore A is a diagonal matrix
we substitute the eigenvalue $\lambda = 1$:

$$\lambda = 1 \implies \left(\begin{array}{ccc|c} 0 & 1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & 1 & 0 & 0 \end{array}\right) \implies \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$x_1 - x_3 = 0 \implies x_1 = x_3$$

$$x_2 - x_3 = 0 \implies x_2 = x_3$$

$$(x_1, x_2, x_3) = (x_3, x_3, x_3) = x_3(1,1,1)$$

for the eigenvalue $\lambda = 1$, the eigenvector is $(1,1,1)$ and the eigenspace is $\text{span}\{(1,1,1)\}$ of dimension 1

Subquestion 10.2. find the diagonal matrix T of A

Answer 10.2.1.

$$T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & -1 & 1 \end{pmatrix}, \quad T^{-1}AT = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Subquestion 10.3. find A^{2020}

Answer 10.3.1.

$$A^{2020} = T \begin{pmatrix} 2^{2020} & 0 & 0 \\ 0 & 2^{2020} & 0 \\ 0 & 0 & 1^{2020} \end{pmatrix} T^{-1}$$

Question 11. given the matrix

$$A = \begin{pmatrix} -2 & 8 & 6 \\ -4 & 10 & 6 \\ 4 & -8 & -4 \end{pmatrix}$$

Subquestion 11.1. find the Characteristic polynomial, the eigenvalues and the eigenvectors of A

Answer 11.1.1.

$$0 = |A - \lambda I| = \left| \begin{pmatrix} -2 & 8 & 6 \\ -4 & 10 & 6 \\ 4 & -8 & -4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = \left| \begin{pmatrix} -2-\lambda & 8 & 6 \\ -4 & 10-\lambda & 6 \\ 4 & -8 & -4-\lambda \end{pmatrix} \right|$$

$$0 = (-2-\lambda)(10-\lambda)(-4-\lambda) + 192 + 192 - (-32(-4\lambda) - 48(-2\lambda) + 24(10-\lambda))$$

$$p(\lambda) = \lambda(\lambda-2)^2$$

the eigenvalues are $\lambda = 2, 2, 0$
we substitute the eigenvalue $\lambda = 2$:

$$\lambda = 2 \implies \left(\begin{array}{ccc|c} -4 & 8 & 6 & 0 \\ -4 & 8 & 6 & 0 \\ 4 & -8 & -6 & 0 \end{array}\right) \implies \left(\begin{array}{ccc|c} 2 & -4 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$x_2, x_3 \text{ free variables}$$

$$x_1 = \frac{4x_2 + 3x_3}{2}$$

$$(x_1, x_2, x_3) = \left(\frac{4x_2 + 3x_3}{2}, x_2, x_3\right) = x_2(2,1,0) + x_3\left(\frac{3}{2}, 0, 1\right)$$

the eigenvalue $\lambda = 2$ has the corresponding eigenvectors $(2,1,0)$, $\left(\frac{3}{2}, 0, 1\right)$ and an eigenspace $\text{span}\{(2,1,0), \left(\frac{3}{2}, 0, 1\right)\}$
we substitute the eigenvalue $\lambda = 0$:

$$\lambda = 0 \implies \begin{pmatrix} -2 & 8 & 6 \\ -4 & 10 & 6 \\ 4 & -8 & -4 \end{pmatrix} \implies \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$x_3 \text{ free variable}$$

$$x_1 = -x_3$$

$$x_2 = -x_3$$

$$(x_1, x_2, x_3) = (-x_3, -x_3, x_3) = x_3(-1, -1, 1)$$

the eigenvalue $\lambda = 2$ has the corresponding eigenvector $(-1,-1,1)$ and an eigenspace $\text{span}\{(-1,-1,1)\}$
and so A is diagonalizable because it has 3 linearly independant eigenvectors and 3 eigenvalues

Subquestion 11.2. find the eigenbasis matrix of A

Answer 11.2.1.

$$T = \begin{pmatrix} 2 & \frac{3}{2} & -1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$T^{-1}AT = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Subquestion 11.3. find A^{2020}

Answer 11.3.1.

$$T^{-1}AT = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$A = T \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} T^{-1}$$

$$A^{2020} = T \begin{pmatrix} 2^{2020} & 0 & 0 \\ 0 & 2^{2020} & 0 \\ 0 & 0 & 0 \end{pmatrix} T^{-1}$$

Question 12. given the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 2 & 7 & 11 \end{pmatrix}$$

Subquestion 12.1. find the characteristic polynomial, the eigenvalues and the eigenvectors of A

Answer 12.1.1.

Step 12.1.1.1. find the characteristic polynomial

$$p(\lambda) = \left| \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 2 & 7 & 11 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = \left| \begin{pmatrix} 1-\lambda & 2 & 1 \\ 0 & 1-\lambda & 3 \\ 2 & 7 & 11-\lambda \end{pmatrix} \right|$$

$$p(\lambda) = -\lambda^3 + 13\lambda^2$$

Step 12.1.1.2. find the eigenvalues

$$0 = -\lambda^3 + 13\lambda^2$$
$$0 = \lambda^2(13 - \lambda)$$
$$\lambda = 13, 0, 0$$

Step 12.1.1.3. find the eigenvectors/eigenspaces
for $\lambda = 13$

$$A - I\lambda = 0$$

$$\left(\begin{array}{ccc|c} 1-13 & 2 & 1 & 0 \\ 0 & -12 & 3 & 0 \\ 2 & 7 & -2 & 0 \end{array}\right) \implies \left(\begin{array}{ccc|c} 1 & 0 & -\frac{1}{8} & 0 \\ 0 & 1 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$x = \frac{1}{8}z$$

$$y = \frac{1}{4}z$$

$$\begin{bmatrix} \frac{z}{8} \\ \frac{z}{4} \\ z \end{bmatrix} \implies z \begin{bmatrix} \frac{1}{8} \\ \frac{1}{4} \\ 1 \end{bmatrix}$$

so the eigenvalue $\lambda = 13$ has the corresponding eigenvector $\begin{bmatrix} \frac{1}{8} \\ \frac{1}{4} \\ 1 \end{bmatrix}$ and an eigenspace span $\left(\left\{\begin{bmatrix} \frac{1}{8} \\ \frac{1}{4} \\ 1 \end{bmatrix}\right\}\right)$

for $\lambda = 0$

$$A - I\lambda = 0$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 2 & 7 & 11 \end{bmatrix} \implies \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x = 5z$$

$$y = -3z$$

$$\begin{bmatrix} 5z \\ -3z \\ z \end{bmatrix} \implies z \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}$$

so the eigenvalue $\lambda = 0$ has the corresponding eigenvector $\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}$ and an eigenspace span $\left(\left\{\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}\right\}\right)$

Subquestion 12.2. find the eigenbasis matrix of A

Answer 12.2.1. A isnt diagonalizable so there isnt an eigenbasis matrix

4.2 linear map composition

remember that linear maps are just functions, so they can be composed

Characteristic 19. let S, T, U be linear spaces, let $f : S \to T, g : T \to U$ be functions, if f and g are linear maps, then also the composite transformation $g \circ f : S \to U$ is a linear map and its transformation matrix is equal to the matrix of f multiplied by the matrix of g correspondingly [**?**]

4.3 linear isomorphism